

Freeze! Financial Sanctions and Bank Responses

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Using regulatory data, we study German bank lending in countries targeted by financial sanctions. We find that domestic banks in Germany reduce lending in sanctioned countries, whereas their foreign bank affiliates outside Germany increase lending. In some cases, this is because the bank affiliates' host countries have not imposed sanctions themselves. However, even German bank affiliates in host countries that enact sanctions like Germany increase lending if these host countries lack strong institutions and anticrime policies. These findings suggest that even universally adopted sanctions distort bank capital flows and competition if the level of their enforcement varies across bank locations. (JEL F51, G18, G28, K3)

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Economic sanctions are an important tool of foreign policy. After World War II, individual states and intergovernmental organizations used especially *nonfinancial* trade restrictions to punish or influence the behavior of targeted entities. However, since the late 1980s, *financial* sanctions, which freeze

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financial assets and investments, have grown in importance. [Felbermayr et al. \(2020\)](#) show that financial sanctions are now imposed more frequently than any other type of economic sanctions, including trade and travel restrictions and arms embargoes.

Financial sanctions are typically targeted at influential individuals, firms or industries that are of strategic or economic importance to the sanctioned country. Banks that operate under the jurisdiction of a sanctioning state must end business with these counterparties and only perform transactions with nontargeted entities in the sanctioned country. Failure to comply can lead to sizeable fines and penalties. For example, in 2015, BNP Paribas was sentenced to pay US\$8.9bn for sanctions violations ([U.S. Department of Justice, 2015](#)). To reduce the risk of litigation, banks can either withdraw business from the sanctioned country or invest in due diligence to verify counterparty information and whether a given transaction remains legal. All else equal, banks should be more likely to withdraw business from sanctioned countries if litigation risk and compliance costs for due diligence are high in the banks' home jurisdictions. Vice versa, banks in laxer jurisdictions could step in and pick up some of the business terminated by other banks. We study this hypothesis using regulatory data on the exposures of German banks with non-German counterparties worldwide.

Previewing our main results, we find that when Germany imposes financial sanctions on behalf of the United Nations (UN) or the European Union (EU), Germany's domestic banks strongly reduce lending in the targeted country.¹ However, German banks' foreign affiliates, subclassified into German bank branches and subsidiaries abroad, on average increase lending relative to their parent banks at home and, in some cases, also in absolute terms. For some of these bank affiliates, the increase in lending can be explained by the fact that the affiliates' host countries do not enact sanctions themselves. However, even some of the German bank affiliates in host countries that do impose sanctions like Germany increase lending. This is particularly true for affiliates in host countries that lack sufficiently strong institutions and anticrime policies to also enforce these sanctions. Overall, our analyses show that German bank business in sanctioned countries partly shifts to German bank affiliates in lax jurisdictions with relatively low litigation risk and compliance costs. One policy implication of this result is that it is not sufficient to ensure that bank host countries all enact sanctions *formally*. Host countries would also need to harmonize their ability or willingness to enforce sanctions to eliminate any distortions to global bank competition favoring banks in lax jurisdictions and to guarantee a universal reduction in lending.

¹ For simplicity, we often refer to bank lending or bank credit. However, for most of our analysis, we use the most comprehensive definition of foreign bank investment and consider all German bank assets with non-German counterparties. In subsample analyses, we find that results are indeed similar for loans and nonloan exposures.

Our empirical analysis makes use of information from the External Position Report of the Deutsche Bundesbank for the years 2002 to 2015. This regulatory data set comprises all German bank assets and liabilities with non-German counterparties. These external positions are disaggregated by destination, but not by an individual counterparty (or by targeted and nontargeted counterparties in sanctioned destinations). For example, we observe, at monthly frequency, a given bank's total outstanding loans with counterparties in Iran. Importantly, all German banks must report their external positions in foreign countries separately. For example, a German bank subsidiary in Panama and its parent bank in Germany would report their exposures in, say, Iran separately from each other. Finally, we observe which banks in Germany and which bank affiliates abroad belong to the same German banking group. Throughout the paper, we follow the terminology of [Fiorentino, Koch, and Rudek \(2010\)](#) and use the term "bank affiliate" as an umbrella term for both bank branches and bank subsidiaries.

We identify the effects of sanctions on different groups of German banks (bank affiliates) in a standard difference-in-differences setting. For example, we compare the business of German banks in Iran before and after the country is targeted by sanctions. Then we examine whether any change in business is different for German banks located in, for example, Germany and Panama. To address concerns that sanctions are not imposed randomly, we control for borrower country fixed effects interacted with month fixed effects ([Khwaja and Mian 2008](#)). These fixed effects control for any macroeconomic and political conditions that could jointly determine aggregate credit demand and the probability of being sanctioned.

In our sample, Germany implements financial sanctions on behalf of the EU or the UN against 13 different borrower countries. After the imposition of sanctions, banks that are domiciled in Germany decrease their positions in targeted countries by, on average, 37%. When we focus on the foreign affiliates of German banks, we find that their positions follow a trend that is parallel to the positions of banks at home in Germany before the imposition of sanctions. However, once sanctions have been implemented, German bank affiliates abroad increase their external positions in the targeted countries by 44% relative to the banks in Germany. This shift in the origin of credit occurs not only across competing German banks but also within German banking groups. Controlling for banking group $\times$ borrower country $\times$ month fixed effects, we observe a relative increase of positions in sanctioned countries that a given banking group manages through its affiliates abroad.

Why does the average German bank affiliate abroad increase its positions relative to the banks at home? Possibly, the difference is explained by affiliates in host countries that do not impose sanctions on behalf of the EU or UN themselves. For this group of affiliates, we find indeed a large incremental effect of 83% relative to banks in Germany. However, even the other German bank affiliates abroad, that is, the affiliates in host countries that are imposing

sanctions themselves, lend relatively more than the banks in Germany. These affiliates still show a positive incremental effect of 35%. The magnitude of these differences in lending across bank locations is economically relevant. Total bank lending aggregated across all targeted countries 12 months before the imposition of EU/UN sanctions amounts to EUR 23.56bn for banks in Germany, EUR 2.59bn for German affiliates in other host countries that impose EU/UN sanctions, and EUR 8.40bn for affiliates in host countries that do not impose EU/UN sanctions. Hence, the incremental growth rates translate into an increase in lending of EUR 0.91bn ($35\% \times \text{EUR } 2.59\text{bn}$) by German bank affiliates in sanctioning host countries and into an increase of EUR 6.97bn by German affiliates in countries that do not impose sanctions. All German affiliates abroad combined show a sizeable incremental effect of EUR 7.88bn, although this amount falls short of the reduction in lending of EUR 8.72bn by the banks located inside Germany.²

Next, we ask why bank lending is not uniform across German banks located in Germany and in other host countries that impose sanctions themselves. One explanation could be that bank host countries that have enacted sanctions formally still differ with respect to their willingness or ability to also enforce these sanctions. For example, some host countries could lack sufficiently strong and well-equipped institutions to persecute and punish sanctions violations, or they could be reluctant to impose policies and regulations that burden banks with complex and costly due diligence, documentation, and reporting requirements. We hypothesize that banks in such host countries will have a competitive advantage for lending in sanctioned countries. This prediction mirrors some ideas in Becker (1968), who theorizes that an individual's decision to commit an offense weighs the expected utility against the expected cost of being apprehended and convicted.

To capture cross-country differences in the quality of institutions and anti-crime policies, we distinguish between bank host countries inside and outside the Financial Action Task Force (FATF). The FATF is an intergovernmental organization that is headquartered at the OECD and sets standards, for example, with respect to client due diligence, to prevent “money laundering, terrorism financing, and other related threats to the integrity of the international financial system.”³ We find that, after Germany has imposed sanctions on behalf of the EU or UN, German bank affiliates outside the FATF increase their positions in the targeted country by, on average, 95% relative to German banks inside the FATF. This incremental effect increases to 151% when we focus on bank affiliates in the nine countries in our sample that the FATF has officially

² Unlike the incremental difference-in-differences estimates for German bank affiliates abroad, the base-level effect of sanctions on lending by the banks at home in Germany lacks a clear counterfactual. Hence, a large part of the EUR 8.72bn decrease in lending by domestic banks ($-37\% \times \text{EUR } 23.56\text{bn}$) could be driven by decreasing credit demand in the sanctioned countries, not by supply-side effects of sanctions.

³ See <http://www.fatf-gafi.org/>.

blacklisted as noncompliant with FATF rules. In several robustness checks, we show that this increase in lending by affiliates in non-FATF countries is not explained by closer economic, cultural, or linguistic ties between non-FATF countries and sanctioned countries. We also show that non-FATF countries support bank lending in sanctioned countries specifically because of weaker *anticrime* policies and institutions and not because of weaker *financial* regulations in general.

We then refine the analysis once more and divide our sample into four subsamples, depending on (a) whether or not a bank's host country acts in a coalition with Germany and enacts EU/UN sanctions at least formally and (b) whether or not the country is a member state of the FATF. We find that only banks in host countries that enact EU/UN sanctions and are also members of the FATF reduce lending in targeted countries. German bank affiliates in other locations increase their positions. Importantly, affiliates in countries with low institutional quality (i.e., affiliates outside the FATF) increase lending in countries targeted by EU/UN sanctions even if their host countries have officially enacted these sanctions themselves.

Clearly, formal enactment of sanctions alone does not ensure a uniform lending response by banks worldwide. This finding has policy implications that are relevant not just in the context of financial sanctions. Bank regulators in many other settings than sanctions (e.g., the Basel framework for prudential bank regulation) seek to harmonize formal rules and laws across countries to eliminate regulatory distortions to international bank competition and to curb regulatory arbitrage. Our analysis suggests that only harmonizing formal rules may not be sufficient and that to ensure a level playing field, countries also need to harmonize their ability or willingness to enforce these rules.

Finally, we examine whether the effect of sanctions on the foreign affiliates of German banking groups depends on their legal status as either branches or subsidiaries. As EU sanctions apply to all business entities that are incorporated under the law of an EU member state (independently of their location), they also apply to German branches outside the EU. Hence, to minimize litigation risk, German banking groups, which are legally liable for their branches worldwide, could prefer to use their foreign subsidiaries rather than their foreign branches for business in sanctioned countries. Consistent with this hypothesis, we show that German bank subsidiaries abroad increase their positions in sanctioned countries by more than bank branches. At the same time, German branches abroad still lend more in sanctioned countries than banks located in Germany. One possible explanation is that branches abroad can perform due diligence and ensure that they only lend to nontargeted counterparties in sanctioned countries at a lower cost than their parent banks. Consistent survey evidence by [LexisNexis \(2020\)](#) shows that due diligence is especially costly in Germany due to high labor costs, strict documentation requirements, and legal limitations on the use of personal data. Another possible explanation could be linked to internal governance problems in

German banking groups. For example, parent banks might not be aware of their branches' lending practices in sanctioned countries.

Our paper is closely related to the literature on international bank regulation. For example, [Dell'Ariccia and Marquez \(2006\)](#) show theoretically how regulatory differences distort competition and put more strictly regulated banks at a relative disadvantage.⁴ Empirically, banking sectors do indeed perform worse in countries with more regulatory restrictions on bank activities ([Barth, Caprio, and Levine 2006](#)). Our work also relates to papers that examine whether banks arbitrage regulatory differences across countries. [Houston, Lin, and Ma \(2012\)](#) find that stronger financial regulations deter inflows of bank funds whereas stronger institutional and legal environments tend to attract them.⁵ We show that a strong institutional and legal environment can also have the opposite effect and reduce banks' competitiveness in markets characterized by legal and compliance risk. This finding is fully consistent with [O'Donovan, Wagner, and Zeume \(2019\)](#), who estimate that 1 in 7 firms shift activities with high legal risk to countries with low institutional quality. More generally, our findings show that harmonizing the enforcement of regulations and laws through effective institutions and policies is as important as harmonizing the rules themselves.

Another related paper in the literature on international regulatory arbitrage is [Karolyi and Taboada \(2015\)](#), who study cross-border bank acquisitions. They find that banks can escape strict regulatory restrictions by acquiring targets in countries with weaker regulations. Building on their work, we show how multinational banks can use their foreign subsidiaries to absorb regulatory shocks by reallocating business internally. Yet another related paper in this literature is [Ongena, Popov, and Udell \(2013\)](#). Focusing on bank lending to foreign counterparties, they show that banks with stricter regulations at home take more *financial risk* abroad (see also [Berrospide et al. 2017](#)), whereas we show that banks in weaker institutional environments take more *nonfinancial risk* (i.e., litigation and compliance risk). To the best of our knowledge, our paper is the first to provide micro-evidence on nonfinancial risk-taking, which, according to [Oliver Wyman \(2018\)](#), generated losses of more than US\$250bn between 2011 and 2016.

At a broader level, our work also contributes to the law, institutions, and finance literature (e.g., [Shleifer and Vishny 1997](#); [La Porta et al. 1998](#); [Beck, Demirgüç-Kunt, and Levine 2003](#); [Acemoglu and Johnson 2005](#)). Most papers in this literature define the quality of law and institutions as the extent to which they protect property rights and enable private contracting. However, a strong legal and institutional framework is not only about promoting private

⁴ Other theoretical papers on international bank regulation include [Acharya \(2003\)](#) and [Morrison and White \(2009\)](#).

⁵ [Demirgüç-Kunt, Horváth, and Huizinga \(2019\)](#) provide consistent evidence showing that local institutional quality determines whether banking groups use the parent or the local subsidiary to lend in a foreign country.

contracting but also about restricting certain transactions. This second aspect is the focus of our paper, which we share with many papers in the literature on crime in finance. For example, [Slutzky, Villamizar-Villegas, and Williams \(2019\)](#) study money laundering on the liability side of banks' balance sheets, whereas we study legal risk on the asset side of banks' balance sheets. Other papers in this literature study terrorism financing ([Limodio 2019](#)), corruption ([Zeume 2017](#); [Giannetti et al. 2021](#)), tax evasion ([Desai, Dyck, and Zingales 2007](#); [Mironov 2013](#); [Beck, Lin, and Ma 2014](#); [Bennedsen and Zeume 2017](#)), misconduct by financial advisers (e.g., [Egan, Matvos, and Seru 2019](#)), or the effect of corporate culture and governance on illicit corporate behavior (e.g., [DeBacker, Heim, and Tran 2015](#); [Guiso, Sapienza, and Zingales 2015](#); [Liu 2016](#)).

Finally, our paper relates to previous research on the economic effects of sanctions on targeted countries. In empirical studies, a wide range of outcome variables are analyzed, including trade ([Haidar 2017](#); [Besedeš, Goldbach, and Nitsch 2022](#)), cross-border financial flows ([Besedeš, Goldbach, and Nitsch 2017, 2021](#)), and economic growth ([Neuenkirch and Neumeier 2015](#)). [Hufbauer et al. \(2007\)](#) and [Biersteker, Tourinho, and Eckert \(2016\)](#) study the design and effectiveness of sanctions as a policy instrument. [Kaempfer and Lowenberg \(2007\)](#) provide an extensive review of the earlier literature on sanctions. Other papers study the relation between geopolitics and trade ([Fuchs and Klann 2013](#)) and capital flows ([Li and Ngo 2017](#)).

1. Institutional Setting and Hypothesis Development

In Section 1.1, we describe the institutional setting of our analysis: (a) what are financial sanctions, (b) which public authorities impose and enforce them, and (c) which companies are legally required to implement them. In Section 1.2, we describe the costs that financial sanctions generate for banks. Section 1.3 derives several hypotheses that guide our empirical analyses.

1.1 Financial sanctions in Germany and the European Union

Sanctions against countries can take various forms, including, for instance, export restrictions, visa and travel bans, and arms embargoes. In our analysis, we focus on financial measures that restrict the free movement of capital through asset freezes. An asset freeze means that targeted entities cannot access existing funds and that no new financial resources or services can be provided to them. Consequently, the restrictions affect the supply of credit, as well as that of other funds and financial services like brokering and international transfer payments or the sale and trade of property. To minimize harm to civilians, policy makers design financial sanctions as “smart sanctions,” with a limited number of targets inside the sanctioned country, while business relations with nontargeted entities remain legal. At the same time, to generate sufficient political leverage, sanctions are typically targeted

at companies and sectors that are of economic relevance and cause spillover effects. Besedeš, Goldbach, and Nitsch (2017) show that financial sanctions indeed reduce the number and value of cross-border financial activities between Germany and sanctioned countries by about 25% to 35%.

In the European Union, the Council of the EU imposes, amends, and lifts sanctions under its Common Foreign and Security Policy (CFSP).⁶ As part of their international commitments, the EU enforces sanctions imposed by the Security Council of the United Nations. In addition, the EU also imposes sanctions autonomously, aiming “to bring about a change in policy or activity by the target country, entities, or individuals” (European Union 2014). In general, the EU member states are responsible for implementing and enforcing all sanctions imposed by the EU Council, that is UN-wide as well as EU-only sanctions. A narrow mandate to coordinate national provisions against sanctions violations and financial crime more broadly was given to the European Banking Authority (EBA) after the money-laundering scandal in an Estonian branch of Danske Bank in 2018 (Arnold and Binham 2019).

EU sanctions apply within the jurisdiction of the EU, that is, within EU territory, to EU nationals, and to companies and organizations incorporated under the law of an EU member state. They apply to EU citizens, companies, and organizations even if these are located outside the EU (European Union 2014). Hence, all branches of EU banks must implement EU sanctions regardless of their location. EU jurisdiction over branches is guaranteed by the fact that the parent banks are legally liable for the activities of their branches. The supervision of a branch outside the EU falls into the responsibility of the EU member state that supervises the parent institute that owns the branch. However, depending on the branches’ host countries, they can be subject to additional local regulations and supervision.⁷ The case of subsidiaries is different because they are generally considered to be legally independent entities. In principle, they are not bound by EU sanctions if they are incorporated outside the EU, although German law requires parent banks to conduct some risk analyses of *group-wide* compliance with antimoney laundering and counterterrorist financing law (GwG, part 2, section 9).

1.2 The costs of financial sanctions for banks

Financial sanctions create two types of costs for banks and other financial institutions that do business in the targeted country. First, if the bank engages in illegal business and breaches sanctions, it faces the risk of future litigation and reputation loss. The largest fine to date (US\$8.9bn) was paid by the French

⁶ See https://www.eeas.europa.eu/eeas/european-union-sanctions_en.

⁷ For example, German authorities also supervise domestic branches and branch offices in Germany if they belong to credit institutions that are domiciled outside Germany (GwG, part 7, section 50). See https://www.bafin.de/SharedDocs/Veroeffentlichungen/EN/Aufsichtsrecht/Gesetz/GwG_en.html?jsessionid=51D252C1B2EEF7E2B2A325BC7004EAD4.1_cid501?nn=8379954#doc7863564bodyText56.

bank BNP Paribas in 2015 (U.S. Department of Justice 2015). Other fines or settlements were paid, among others, by British, Danish, Dutch, French, German, and Italian banks. Between 2012 and 2018, European banks were fined over US\$16bn for money laundering and sanction breaches (Moody's 2019). Hence, litigation costs are significant. However, from the (ex ante) perspective of a bank that lends or otherwise trades in sanctioned countries, they need to be weighed with the probability of conviction. More generally, expected litigation costs depend on "the amount and type of resources and punishments used to enforce a piece of legislation" (Becker 1968, p. 169), which vary across banks' different host countries.

Second, sanctions also make bank business more costly because of additional compliance requirements with respect to regulatory reporting and client due diligence. These costs can also make legal business with nontargeted entities in the sanctioned countries more expensive. In a global survey conducted by LexisNexis (2020), roughly half of the 898 polled banks, insurance companies, and investment and asset management firms answered that screening for sanctions violations and customer profiling regarding financial crime were key challenges in compliance management. Roughly 40% of the firms identified regulatory reporting on financial crime as a major challenge. Half of the surveyed firms reported a negative impact of compliance processes on their productivity and customer acquisition; two out of three firms were concerned about negative effects on employee satisfaction. In 2020, financial institutions worldwide spent an estimated US\$180.9bn on compliance processes related to financial crime.

The costs of financial crime compliance vary sizably across countries. According to the survey by LexisNexis (2020), European financial firms represent the largest portion (US\$136.5bn) of the total US\$180.9bn spent worldwide, with Germany alone accounting for US\$47.5bn. The average German financial firm with assets above US\$10bn reported compliance costs of US\$46.1m in 2020 compared to an average of only US\$24.7m in the full sample of large firms from all countries in the survey. Stricter documentation requirements, high labor costs, legal limitations on the use of personal data, and, as a result, longer due diligence times are responsible for the higher compliance costs in the EU and especially in Germany. In 2020, European financial firms spent 47 hours on due diligence per foreign client compared to 25 hours spent by North American firms and a global average of 34 hours.

1.3 Hypotheses

We formulate three hypotheses that predict the credit supply of German banks. Following Becker (1968), we assume that a bank's decision to lend in sanctioned countries depends on a trade-off between expected profits and sanctions-related costs. All else equal, an increase in (expected) litigation and compliance costs following the imposition of sanctions should decrease

the supply of credit to the targeted country. However, as we do not observe litigation and compliance costs directly in our data, we assume that they depend on a bank's institutional environment, which varies across banks' different host countries.

Hypothesis 1: The effect of financial sanctions on lending across bank locations

Banks that lend in sanctioned countries incur litigation and compliance costs, whose size depends on the local institutional quality in banks' host countries. After sanctions are imposed, banks in locations with weak anticrime policies and law enforcement increase their positions in the sanctioned country relative to other banks.

As explained in Section 1.1, financial sanctions do not only restrict lending but also freeze existing deposits of targeted counterparties.⁸ The lending channel predicts that banks located in host countries with low institutional quality and weak anticrime policies lend relatively more in sanctioned countries (Hypothesis 1), whereas the deposit channel predicts that banks in weak jurisdictions freeze relatively less deposits and banks pay out relatively more cash. In other words, both channels make opposite predictions for banks' balance sheet size. We focus on the lending channel and shut down the deposit channel empirically as our dependent variable is not banks' total balance sheet but only banks' external positions abroad excluding cash. We also implement robustness tests that exclude cash-like positions with foreign counterparties (e.g., money market papers) and only consider loans (as opposed to all noncash positions) in sanctioned countries (see Internet Appendix 4, Table A.1). This robustness test alleviates concerns that banks do not meet their depositors' cash withdrawals with reductions in their own cash holdings but by selling liquid assets.

We also note that Hypothesis 1 is not limited to German banks and likely applies to banks of other nationalities, too. Weak anticrime policies and law enforcement should provide all banks in relatively lax jurisdictions with a competitive advantage for business in sanctioned countries. Hence, we would expect that two mechanisms are at work simultaneously. First, German credit supply in sanctioned countries should shift from German banks located in strict, high-cost jurisdictions toward the affiliates of German banking groups that are located in lax, low-cost jurisdictions. In addition to such a geographic shift in German credit supply, a second shift should take place also across banking groups of different nationalities as non-German banks in lax jurisdictions pick up some of the lending business that their (German) competitors in stricter jurisdictions terminated. As our data do not include banks of other nationalities, we restrict our empirical analysis to German positions.

⁸ We thank an anonymous associate editor for raising this point.

Hypothesis 1 implies that financial sanctions distort international bank competition by giving banks in lax host countries a cost advantage for business in the sanctioned country. Conceptually, this prediction is related to a fundamental challenge to the design of international financial architecture: cross-country differences in financial regulation can distort competition, because finance is a global industry dominated by multinational banks, whereas regulation is typically national (Barth, Caprio, and Levine 2004, 2013). A common response to this problem is to harmonize regulations across countries. We predict that harmonizing the laws and regulations themselves is necessary, but not sufficient. Competitive distortions will likely persist unless countries also harmonize the *enforcement* of these rules. To test this prediction, we compare the activities of banks that operate from host countries that all *formally* enact sanctions, but that differ with respect to the quality of their institutions and anticrime policies and, hence, their ability or willingness to enforce the sanctions.

Hypothesis 2: Formal enactment versus enforcement of financial sanctions

After Germany imposes sanctions against a given counterparty country on behalf of the EU or UN, German banks in host countries that either (a) do not enact these sanctions themselves or (b) enact them formally but lack strong institutions and anticrime policies to enforce them increase lending relative to other German banks.

Finally, we predict that branches and subsidiaries of German banking groups respond differently to the imposition of financial sanctions. As explained in Section 1.1, German banks are legally liable for their branches worldwide whereas their subsidiaries are generally considered as legally independent entities. Furthermore, German branches are incorporated under German law and are, hence, required to respect the sanctions that Germany enacts on behalf of the EU or UN. Hence, it seems likely that, to reduce litigation risk, German banking groups are more reluctant to use their foreign branches than their subsidiaries for business in sanctioned countries. Further, if German banks monitor their branches (more than their subsidiaries) internally, such group-internal control could compensate for weak external monitoring by local institutions in a branch's host country. Overall, we predict that bank branches in locations with weak institutional quality increase their business in sanctioned countries less aggressively than the foreign subsidiaries of German banking groups.

Hypothesis 3: The effects of financial sanctions on bank branches versus subsidiaries

Following the imposition of sanctions, the shift in the origin of German bank lending toward bank locations with weak local institutions and anticrime policies is stronger for subsidiaries than for branches of German banking groups.

2. Data

In this Section, we explain how we measure the institutional environment in the different host countries of German banks and describe our data on banks' external positions.

2.1 Measuring institutional quality across bank locations

German banks that lend in sanctioned countries incur litigation and compliance costs. We approximate these costs by a bank's institutional environment in terms of regulation, supervision, and enforcement of financial sanctions. We use three alternative approaches to distinguish between German banks in countries with high and with low institutional quality.

Our first approach compares banks that are located in Germany to German bank affiliates that are located abroad. This categorization has several advantages. First, it does not rely on any subjective criteria of how to evaluate the different dimensions of institutional quality. Instead, it answers the basic question of whether German banking groups prefer to use their affiliates abroad to lend in sanctioned countries. Second, Germany (along with the United Kingdom) tops the list of countries with the highest compliance costs for financial crime, according to [LexisNexis \(2020\)](#). Hence, a comparison between banks inside and outside of Germany provides a useful test for the hypotheses developed in the previous section. Finally, the distinction between banks in Germany and abroad offers the econometric advantage that both categories have a similar number of observations in our sample and their total amount of lending does not differ too much, either (see Sections 2.3 and 2.4).

The main limitation of a simple distinction between Germany and the rest of the world as host locations is that, by doing so, we effectively pool foreign countries with strong and weak policies against financial crime. The resultant classification or measurement error will bias our analysis against finding evidence for our hypotheses. Therefore, our second proxy of institutional quality aims to alleviate this shortcoming. Specifically, we group bank locations by countries' membership status in the FATF. The FATF is a large intergovernmental organization of developed democracies and was founded on the initiative of the G7 in 1989. Its explicit objective is to combat "money laundering, terrorist financing, and other related threats to the integrity of the international financial system."⁹ The FATF asks its member states: "to freeze without delay the funds or other assets of, and to ensure that no funds or other assets are made available, directly or indirectly, to or for the benefit of, any person or entity either (i) designated by, or under the authority of, the United Nations Security Council under Chapter VII of the Charter of the United Nations" ([FATF 2012](#), p. 11). The FATF sets explicit standards regarding, for

⁹ See <http://www.fatf-gafi.org/>.

example, customer due diligence, recordkeeping, transparency of counterparty information, and business in high-risk countries.

FATF membership is a good proxy for a country's commitment to enforce sanctions. Still, some limitations remain. For instance, besides committing to FATF standards, a country must also have a sufficiently large and developed financial sector to become a FATF member. As a result, some smaller countries might not be members in the FATF but participate in regional organizations with similar objectives.¹⁰ Moreover, some FATF members might have become partly noncompliant over time and, like Russia, might even be subject to sanctions today.¹¹ As before, this sort of classification error will bias our analyses against finding evidence for our hypotheses.

Our third classification approach is the most stringent in identifying bank locations with weak policies against financial crime and a lack of commitment to financial sanctions. It exploits the fact that the FATF periodically audits member states as well as nonmember states. Countries that do not comply with FATF standards are declared noncooperative. Our sample comprises nine noncooperative countries that host German bank affiliates: Cyprus, Guernsey, Iran, Jersey, Malta, Mauritius, Pakistan, the Philippines, and Russia. The small number of countries is the main disadvantage of focusing on FATF-blacklisted countries. Overall, when choosing between our different approaches to classifying banks by location, we trade-off measurement error from misclassification against small group size. Therefore, to demonstrate the robustness of our analysis, we report results for all three classifications, without particular emphasis on one of these approaches.

2.2 Data sources

Our main data source is the Deutsche Bundesbank's External Position Report. This report, which the Deutsche Bundesbank regularly compiles based on banks' mandatory reporting, contains granular information about the assets and liabilities vis-à-vis nonresidents of the entire German banking population (roughly 2,000 banks), including German banks located in Germany, their subsidiaries and branches abroad, as well as the subsidiaries of foreign banks operating in Germany. Most notably, the external positions of subsidiaries and branches are not attributed to their German parent banks but recorded separately.¹² Fiorentino, Koch, and Rudek (2010) and Krueger, Munzert, and Stahl (2017) provide a detailed description of this data set. The data are available to external researchers only on the premises of the central bank. For

¹⁰ For example, whereas Argentina and Brazil are the only South American members of the FATF, most other South American countries are members of the Grupo de Acción Financiera de Latinoamérica (GAFILAT).

¹¹ Figure A.1 in Internet Appendix 4 shows FATF members as of October 2017. Many (offshore) financial centers (e.g., Cayman Islands, Isle of Man, Jersey, Liechtenstein, Malta, Monaco) are not members.

¹² Each subsidiary files its own report. Branches of the same group in the same country submit a joint report.

confidentiality reasons, the Deutsche Bundesbank anonymizes bank names and randomly deletes 10% of the positions.¹³

The External Position Report contains all assets and liabilities with foreign counterparties as well as domestically held assets and liabilities denominated in foreign currency on a monthly basis since March 2002. Positions in 2002 are only recorded for institutions with external positions above Euro 10m. Yet, with this threshold level, the data still cover more than 90% of total volume. Since January 2003, literally all positions must be reported, and the database also includes positions for which external assets have dropped to zero (roughly 25% of all observations in the sample). Table A.2 in Internet Appendix 4 shows that our main results are robust to excluding the year 2002 from our analysis. One observation in the data set is a bank-country-month triplet (b, c, t) in which c is one given destination country of bank b in month t .¹⁴ Hence, individual positions of bank b with counterparties in country c are aggregated, and counterparty identities are not observed. The aggregation takes place within several categories for different types of counterparties (households, nonfinancial corporations, banks, etc.), asset classes, and currencies.

We match these data with public information on the financial sanctions that Germany has imposed between 2002 and 2015 on behalf of the EU or UN. The information is mainly obtained from the service center Financial Sanctions of the Deutsche Bundesbank, which is responsible for the enforcement of financial sanctions under EU regulations in Germany.¹⁵ We augment these data with information from official EU sources.¹⁶ Table 1 lists the 13 counterparty countries that are subject to EU/UN financial sanctions during our sample period. Of these, only eleven change their status as sanctioned countries as Iraq and Myanmar were already sanctioned before 2002. Finally, we use information from the Global Sanctions Database (GSDB) to determine which host countries of German bank affiliates also enact the EU/UN sanctions that Germany imposes against a given target country.¹⁷ The GSDB reports financial sanctions only at annual frequency. We merge these data with our monthly data sets based on overlaps in the 13 sanctions episodes listed in Table 1.¹⁸

¹³ The External Position Report is used by the banking supervisors, the balance of payments analysis division, and the monetary analysis division of the Deutsche Bundesbank. Further recipients of (aggregated) data are the European Central Bank (ECB) and the Bank for International Settlements (BIS).

¹⁴ Destination countries include all sovereign countries, as well as several offshore destinations, that must be reported explicitly.

¹⁵ See <https://www.bundesbank.de/en/service/financial-sanctions/financial-sanctions-619000>.

¹⁶ Common Foreign and Security Policy Decisions and European Union Regulations are published in the Official Journal of the EU; see <http://eur-lex.europa.eu/homepage.html>.

¹⁷ Felbermayr et al. (2020) describe the GSDB in detail.

¹⁸ To merge the GSDB with our main data set, we proceed as follows. For each of the sanctions episodes in Table 1, we use the GSDB to identify the host countries of German banks that are in a coalition with Germany and impose the EU/UN financial sanctions themselves. As the GSDB reports at annual frequency but our main data are monthly, we assume that these host countries start and end the sanctions in the same months as Germany. For

Table 1
List of financial sanctions newly imposed between 2002 and 2015

	Announcement (1)	Additional nonfinancial sanctions (2)	Imposed by (3)
Iraq	Dec. 9, 1992	Restrictions of contracts and business activities	Entire UN
Myanmar	May 22, 2000	Export restriction on military equipment	Only EU
Liberia	Sept. 4, 2003 (lifted Jun. 20, 2016)	Export restriction on military equipment	Entire UN
Lebanon	Feb. 21, 2006	–	Entire UN
Belarus	May 18, 2006	Export restriction on military equipment	Only EU
Iran	Feb. 2, 2007	Export restriction on military equipment, chemicals, and other resources	Entire UN
Tunisia	Feb. 4, 2011	–	Only EU
Libya	Mar. 2, 2011	Export restriction on military equipment	Entire UN
Egypt	Mar. 21, 2011	–	Only EU
Syria	May 9, 2011	Export restriction on military equipment, chemicals, and other resources	Only EU
Afghanistan	Aug. 1, 2011	–	Entire UN
Russia	Mar. 5, 2014	Export restriction on oil drilling machinery, chemicals, and other resources	Only EU
Yemen	Dec. 18, 2014	–	Entire UN

This table lists the 13 countries in our sample that are sanctioned in at least 1 month between 2002 and 2015 and in which at least one German bank owns a position. *Sources:* Deutsche Bundesbank, Service center “Financial Sanctions,” and authors’ own investigations.

2.3 Summary statistics

Panel A of Table 2 provides summary statistics for the 2,390,051 bank-counterparty country-month triplets (b, c, t) in our sample. German bank affiliates located outside Germany account for 49% of the sample. Eighteen percent and eight percent of all observations belong to bank affiliates that are located outside the EU and outside the FATF, respectively. Two percent of all observations belong to positions between a German affiliate and a non-German counterparty that are both located in the same country. An example would be the subsidiary of a German bank in France that reports its positions with French entities. Business relations with counterparties in the 13 countries sanctioned by Germany on behalf of the EU or UN (see Table 1) account for 67,628 observations, or 2.8% of the sample. Of these 67,628 observations, 60,514 (i.e., 2.5% of the full sample) belong to banks in Germany or to German bank affiliates in other host countries that also enact the sanctions listed in Table 1. The remaining 7,114 observations belong to bank affiliates in host countries that do not enact these sanctions that Germany imposes on behalf of the EU/UN.

The average bank-country-month triplet (b, c, t) has total external assets (excluding banks’ cash holdings) of EUR 231,440,790 (aggregated across all

instance, when Japan imposed financial sanctions against Russia in 2014, we assume that Japan started these sanctions in March 2014, that is, in the same month as Germany. In a few rare cases, a host country imposes financial sanctions against one of the targets in Table 1 only one or several years after Germany. For example, according to the GSDB, the United States imposed financial sanctions against Egypt in 2013 only, whereas German financial sanctions have been in place since March 2011. In this case, we assume that the United States start sanctions in January 2013.

Table 2
Position-level summary statistics

A. Full sample

	Obs. (1)	Mean (2)	SD (3)	Skew. (4)
Host country indicators of German bank:				
<i>Abroad_b</i>	2,390,051	0.49	0.50	
<i>Outside EU_b</i>	2,390,051	0.18	0.39	
<i>Outside FATF_b</i>	2,390,051	0.08	0.26	
<i>In counterp. country_b</i>	2,390,051	0.02	0.15	
Sanction indicators:				
<i>EU/UN sanctions_{c,t}</i>	2,390,051	0.028	0.166	
<i>Bank host country enacts</i>	2,390,051	0.025	0.157	
<i>EU/UN sanctions_{b,c,t}</i>				
External assets <i>Assets_{b,c,t}</i> by counterparty (in EUR 1,000):				
All counterparty types	2,390,051	231,440.79	2,510,851.22	44.86
Banks	2,390,051	96,146.07	1,308,037.50	47.59
Other financials	2,390,051	9,056.44	301,561.90	96.76
Government	2,390,051	3,434.71	66,409.87	49.34
Nonfinancial firms, households, nonprofit	2,390,051	61,873.03	1,181,520.67	68.74
External assets <i>Assets_{b,c,t}</i> by asset class (in EUR 1,000):				
Loans	2,390,051	170,510.24	2,129,962.98	45.63
Nonloan assets	2,390,051	60,930.55	567,338.02	40.74
Treasury bills	2,390,051	698.19	30,171.74	106.06
Money market funds	2,390,051	2,796.53	84,405.42	11.81
Fixed-income securities	2,390,051	45,446.67	352,945.25	29.06
Shares	2,390,051	5,683.31	247,760.77	95.95
External assets <i>Assets_{b,c,t}</i> by host country (in EUR 1,000):				
Banks inside DEU	1,220,100	234,620.10	1,962,711	34.17
Banks abroad	1,169,951	228,125.20	2,976,845	44.78
Banks outside EU	438,153	223,229.60	2,498,737	29.29
Banks outside FATF	180,446	195,182.80	2,828,796	31.97
<i>Log (total) assets_{b,c,t}</i>	2,386,762	5.92	4.85	0.13

(Continued)

foreign counterparties and asset classes). Among the different counterparties, business with banks, including intragroup positions between bank *b* and its affiliates in country *c*, are the largest position and amount to, on average, EUR 96,146,070.¹⁹ Positions with foreign nonfinancial corporations, households, and nonprofit institutions have an average size of EUR 61,873,030. The largest asset class are loans, which, on average, equal EUR 170,510,240, whereas all nonloan positions are, on average, worth EUR 60,930,550. Banks located in Germany hold the largest external positions with an average size of EUR 234,620,100. However, German banks abroad hold only marginally smaller positions of, on average, EUR 228,125,200. All distributions shown in panel

¹⁹ It is not possible to separate the external positions with nonaffiliated banks from the external positions with affiliated branches and subsidiaries as reporting banks do not disclose the identities of individual counterparties. We discuss this limitation in detail in Internet Appendix 1. Our results are robust if we exclude positions with financial counterparties (i.e., if we exclude positions with affiliated and nonaffiliated banks) in Table A.3.

Table 2
(Continued)*B. Sanctioned versus never sanctioned counterparty countries*

	Sanctioned at least once			Never sanctioned		
	Obs. (1)	Mean (2)	SD (3)	Obs. (4)	Mean (5)	SD (6)
Host country indicators of German bank:						
<i>Abroad_b</i>	159,107	0.43	0.49	2,230,944	0.49	0.50
<i>Outside EU_b</i>	159,107	0.15	0.35	2,230,944	0.19	0.39
<i>Outside FATF_b</i>	159,107	0.07	0.26	2,230,944	0.08	0.26
<i>In counterp. country_b</i>	159,107	0.01	0.07	2,230,944	0.02	0.15
External assets <i>Assets_{b,c,t}</i> by counterparty (in EUR 1,000):						
All counterparty types	159,107	37,118.39	268,686.54	2,230,944	245,299.52	2,597,298.05
Banks	159,107	14,480.36	148,378.98	2,230,944	101,970.32	1,353,109.47
Other financials	159,107	112.92	2,938.54	2,230,944	9,694.27	312,119.37
Government	159,107	1,139.94	24,179.01	2,230,944	3,598.37	68,430.31
Nonfinancial firms, households, nonprofit	159,107	17,995.81	112,783.02	2,230,944	65,002.27	1,222,496.02
External assets <i>Assets_{b,c,t}</i> by asset class (in EUR 1,000):						
Loans	159,107	33,729.03	218,300.31	2,230,944	180,265.23	2,203,512.38
Nonloan assets, i.e.,	159,107	3,389.36	84,019.39	2,230,944	65,034.28	586,575.99
Treasury bills	159,107	96.32	6,050.41	2,230,944	741.11	31,186.84
Fixed-income securities	159,107	1,879.82	52,433.59	2,230,944	48,553.78	364,847.09
Shares	159,107	895.83	35,198.47	2,230,944	6,024.74	256,267.81
External assets <i>Assets_{b,c,t}</i> by host country (in EUR 1,000):						
Banks inside DEU	90,819	41,081.21	231,612	1,129,281	250,184.80	2,038,252
Banks abroad	68,288	31,848.07	311,149	1,101,663	240,291.70	3,066,328
Banks outside EU	23,096	24,355.77	163,979	415,057	234,294.90	2,566,573
Banks outside FATF	11,228	4,837.94	19,427	169,218	207,811.60	2,920,695
<i>Log (total) assets_{b,c,t}</i>	158,908	3.69	4.29	2,227,854	6.08	4.85

This table provides summary statistics for the external positions in 105 foreign countries of 192 domestic banks in Germany plus their affiliated branches and subsidiaries abroad. One observation is a bank-counterparty country-month triplet (b, c, t). Panel A shows variables describing the host country of bank b , whether the country of counterparty is sanctioned, and external positions disaggregated by counterparty type and asset class in the full sample. Panel B shows the same variables separately for positions in the 13 sanctioned countries listed in Table 1 (columns 1 to 3) and all other positions (columns 4 to 6). All positions are reported in EUR 1,000. The sample is a random extraction of Deutsche Bundesbank's External Position Report and covers 90% of all external positions of all German banks worldwide between March 2002 and December 2015. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and authors' own calculations.

A exhibit large sample variation and long right tails.²⁰ To reduce the influence of extreme realizations, we calculate the natural logarithm of one plus total external bank assets $Assets_{b,c,t}$, which reduces the skewness from 44.86 to 0.13. Our regression estimates are qualitatively and quantitatively robust if, instead, we calculate the inverse hyperbolic sine transformation of $Assets_{b,c,t}$ (see Internet Appendix 4, Table A.4).

Panel B of Table 2 presents separate summary statistics for positions in sanctioned countries (columns 1 to 3) and for positions in all other countries (columns 4 to 6). In both groups, the distribution of bank locations is very similar. German bank affiliates located outside Germany account for 49% and

²⁰ For confidentiality reasons, we are unable to report individual data points like minima and maxima.

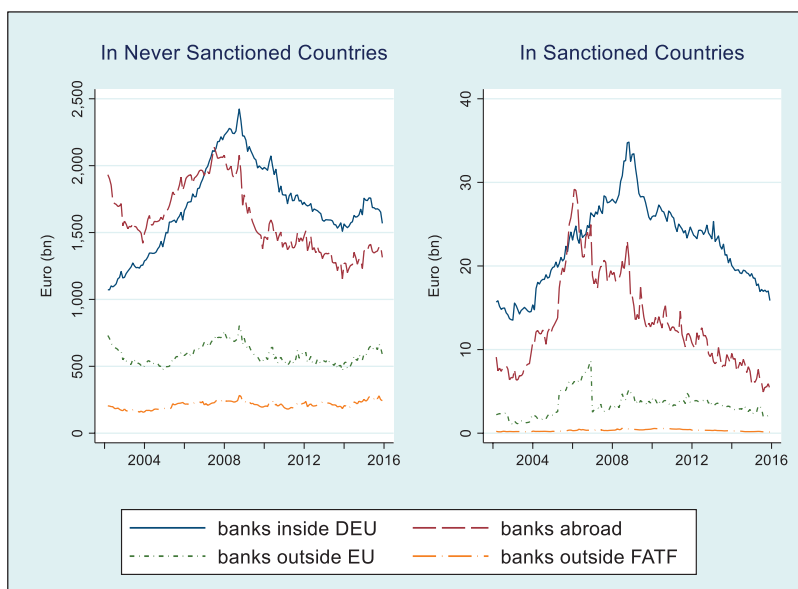


Figure 1

External positions of German banks This figure shows the evolution of total external positions owned by German banks with counterparties in countries that have never been sanctioned (left-hand graph) or have been sanctioned at least once between 2002 and 2015 (right-hand graph). Positions are aggregated over all counterparties, asset classes, and currencies of denomination. They are broken down into total external assets of all banks located inside Germany (DEU), of all German bank affiliates abroad, and of the subset of German bank affiliates that are located outside the EU or outside the FATF. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations.

43% of all positions in never-sanctioned and sanctioned countries, respectively. Banks in Germany hold positions in sanctioned countries of, on average, EUR 41,081,210, whereas their affiliates outside Germany hold positions in sanctioned countries of, on average, EUR 31,848,070. Unsurprisingly, positions in countries that are never sanctioned tend to be much larger.

2.4 Geographic distribution and evolution of banks' cross-border activity

Figure 1 plots the time series of German banks' external positions. The graph on the left-hand side shows positions with counterparties in countries that are never sanctioned between 2002 and 2015. German bank affiliates abroad hold almost as many foreign assets as their parent institutions at home. The subset of affiliates outside the EU accounts for roughly one-third of all external positions. German bank affiliates outside the FATF own relatively small but still significant positions with foreign counterparties of about EUR 250bn. External positions grew rapidly until 2009 and declined again afterward. The graph on right-hand side of Figure 1 provides analogous plots for external positions with

Table 3
Country-level summary statistics

	Obs. (1)	Mean (2)	SD (3)	Skew. (4)	P10 (5)	P50 (6)	P90 (7)
<i>A. All counterparty countries</i>							
Aggregated Assets _{c,t} (EUR bn)	17,008	32.50	100.00	5.68	0.12	3.27	66.70
Asset growth rate _{c,t}	16,851	0.159	6.743	72.28	-0.101	0.000	0.111
Volatility (asset growth rate) _c	105	1.072	7.514	9.64	0.045	0.098	0.431
<i>B. Never-sanctioned countries</i>							
Aggregated Assets _{c,t} (EUR bn)	14,850	36.90	107.00	5.30	0.27	4.05	93.60
Asset growth rate _{c,t}	14,715	0.147	7.060	71.56	-0.093	0.000	0.103
Volatility (asset growth rate) _c	92	0.965	7.917	9.43	0.048	0.091	0.270
<i>C. Sanctioned countries</i>							
Aggregated Assets _{c,t} (EUR bn)							
All periods	2,158	2.74	7.29	3.60	0.00	0.18	5.13
Before imposition of sanctions	1,021	4.29	9.98	2.43	0.00	0.15	22.60
After imposition of sanctions	1,137	1.34	2.67	4.20	0.00	0.38	4.00
Asset growth rate _{c,t}	2,136	0.243	3.920	23.36	-0.201	0.000	0.246
Volatility (asset growth rate) _c							
All assets	13	1.833	3.646	1.91	0.045	0.369	9.332
Only loans	13	1.938	3.620	1.87	0.043	0.499	9.332
Only nonloan assets	13	36.071	55.387	1.22	0.323	4.324	146.47

This table reports summary statistics for aggregated bank assets (in EUR bn) in 105 counterparty countries. One observation is a counterparty country-month pair (c, t). We further report the monthly growth rate of aggregated bank assets and the volatility of asset growth calculated over 2002 and 2015 at the country level. Panel A shows summary statistics for the full sample of all 105 counterparty countries. Panels B and C show separate statistics for the 92 counterparty countries that are never sanctioned between 2002 and 2015 and for the 13 countries that are sanctioned at least once during our sample period. Panel C further reports separate statistics for aggregated bank positions before and after the imposition of financial sanctions. Aggregated positions are calculated based on a random extraction of Deutsche Bundesbank's External Position Report that covers 90% of all external positions of all German banks worldwide between March 2002 and December 2015. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations.

counterparties in countries that were sanctioned at least once between 2002 and 2015. Again, assets owned by bank affiliates abroad account for an important share of total German positions.

Figure 1 shows that banks' positions in sanctioned countries are much smaller than those in countries that are never sanctioned between 2002 and 2015. Table 3 provides more detailed information regarding the distribution of these positions. In the group of never-sanctioned countries, average German bank assets, aggregated at the country level, equal EUR 36.9bn (panel B of Table 3). By contrast, counterparties in countries that are sanctioned in at least 1 month (panel C) owe the German banking sector, on average, only EUR 2.74bn. Panel C further shows that German bank assets in sanctioned countries decrease from, on average, EUR 4.29bn before the imposition of the sanctions to EUR 1.34bn afterward. The distribution of German bank assets is positively skewed for sanctioned as well as never-sanctioned countries. Sample medians are much smaller than sample averages and values at the 90th percentile are large.

Analyzing changes in German bank assets at the monthly frequency, we find a median growth rate of 0.000 for sanctioned as well as never-sanctioned countries. However, for sanctioned countries, 10% of the observations show a negative growth rate of less than -0.201 (panel C), compared to only -0.093 for never-sanctioned countries (panel B). Finally, positions in sanctioned countries seem to be more volatile. For the average sanctioned country, asset growth rates have a volatility of 1.833 compared to only 0.965 for the average never-sanctioned country. German loans in sanctioned countries appear to be less volatile compared to nonloan positions (e.g., tradable securities).

Finally, we note that the global outreach of the German banking sector also manifests itself in a large global network of German bank affiliates. At the start of our sample in 2002, about 80 German banking groups have at least one affiliated establishment abroad. Roughly 30 of them have a bank affiliate outside the EU and about 20 outside the FATF. These branches and subsidiaries are located in more than 60 different host countries, of which a little less than 40 (respectively, 30) are located outside the EU (FATF). These numbers decrease somewhat over time but remain high. To preserve space, we show the details of this evolution in [Internet Appendix 4 \(Figure A.2\)](#).

3. Methodology

In this section, we describe the empirical test for our main prediction (Hypothesis 1). We examine the effect of financial sanctions on the external position of bank b with counterparties in country c in month t in a standard differences-in-differences setting:

$$\begin{aligned} \text{Log } assets_{b,c,t} = & \beta_S EU/UN \text{ sanctions}_{c,t} + \beta_T EU/UN \text{ sanctions}_{c,t} \\ & \times \text{Weak inst'l quality}_b + \alpha_{b,c} + \alpha_{b,t} + \varepsilon_{b,c,t}, \end{aligned} \quad (1)$$

where we use total external assets (including nonloan positions but excluding cash holdings) with counterparties in country c as dependent variable because it is the most comprehensive measure of cross-border bank investment.²¹ $EU/UN \text{ sanctions}_{c,t}$ is an indicator variable that equals one if Germany has imposed financial sanctions against counterparty country c in month t on behalf of the EU or UN and zero otherwise. $\text{Weak inst'l quality}_b$ is an indicator variable that takes the value of one if bank b is located in a host country with presumably weak institutional quality (and zero otherwise). We include bank $\times$ time ($\alpha_{b,t}$) and counterparty country $\times$ bank fixed effects ($\alpha_{b,c}$). In keeping with Hypothesis 1, we expect a positive coefficient $\hat{\beta}_T$ for the interaction term.

²¹ Our results are robust when we only consider bank loans and exclude all other assets like, for example, money market papers and other cash-like assets (see [Internet Appendix 4, Table A.1](#)).

We acknowledge three identification challenges to our baseline specification. The first one is that sanctions are not imposed randomly but in response to political developments in the target country (war, violations of human rights or UN resolutions, etc.). These underlying political developments as well as the sanctions themselves have direct effects on macroeconomic fundamentals and can therefore shift the demand for credit in the targeted country. To control for changes in credit demand, we estimate additional regressions that control for counterparty country fixed effects interacted with time fixed effects ($\alpha_{c,t}$).

A second identification challenge arises because the institutional quality of bank locations is not exogenous but correlated with other characteristics of banks' host countries. For example, it could be the case that bank locations with weak institutions have special cultural or economic ties with sanctioned countries. Also, bank locations with weak institutions probably have weaker financial standards and lower regulatory costs in general, not just with regard to financial sanctions. Although the bank $\times$ counterparty country fixed effects $\alpha_{b,c}$ in Equation (1) capture all time-invariant differences across bank locations and even across bank-counterparty country pairs, they do not control for the possibility that persistent differences across bank locations condition the effect of sanctions itself. In the robustness section, we discuss specific examples for this concern. To control for alternative channels through which sanctions affect the geographic composition of bank lending, we augment Equation (1) with additional interaction terms $EU/UN\ sanctions_{c,t} \times Bank\ location\ characteristics_b$ in Section 5.

A third identification challenge arises if banks adjust their supply of credit due to events that correlate with the imposition of financial sanctions. For example, Iran was sanctioned in 2007 and, therefore, only months before the onset of the financial crisis, which likely had heterogeneous effects on different banks. The bank $\times$ time fixed effects $\alpha_{b,t}$ in Equation (1) help to the extent that they control for time variation in unobserved bank-level variables like bank-wide credit supply aggregated across all counterparty countries. However, they do not control for the possibility that, for example, banking groups have counterparty country-specific credit policies and could, hence, react differently to the imposition of financial sanctions. Therefore, we also estimate additional regressions with banking group-counterparty country $\times$ month fixed effects ($\alpha_{g,c,t}$). In those specifications, identification only comes from the within-group allocation of business in sanctioned countries and not from a shift in credit supply across competing banks.

We compute three-way standard errors. We cluster by bank location and by counterparty country because banks and, respectively, counterparties that reside in the same country are exposed to the same local macroeconomic shocks. Hence, we expect banks' supply and, respectively, counterparties' demand of credit to be correlated within countries. Finally, we cluster by month to allow the regression residuals to be correlated over time.

4. Empirical Analysis

Our empirical analyses match the hypotheses stated in Section 1.3. In Section 4.1, we show that, after the imposition of financial sanctions, German banks in locations with weak institutional quality in terms of regulation, supervision, and law enforcement increase their positions in targeted countries relative to German banks in host countries with stronger institutions. In Section 4.2, we show evidence that, to avoid distortions to international bank competition, it is not sufficient to harmonize the formal enactment of sanctions across banks' different host countries but that the enforcement of sanctions through strong institutions and policies must be harmonized as well. In Section 4.3, we show that financial sanctions have a different effect on the lending decisions of German bank branches versus subsidiaries.

4.1 The effects of financial sanctions on lending across bank locations

In Table 4, column 1, we show a specification of Equation (1) with the variable $EU/UN\ sanctions_{c,t}$ but without any interaction terms. The coefficient of -0.275 suggests that German bank positions with foreign counterparties decrease by, on average, 24% ($=\exp(-0.275)-1$) after Germany imposes financial sanctions on behalf of the EU or UN. There are two possible explanations for this result. First, German banks supply less credit in sanctioned countries as sanctions increase the litigation and compliance cost for business. Second, sanctions and the underlying political developments depress economic activity and, hence, the demand for credit.

Columns 2 to 10 show regressions with interaction terms between $EU/UN\ sanctions_{c,t}$ and our three different proxies that identify bank locations with relatively weak institutional quality. For the reasons explained in Section 2.1, the indicator variable $Abroad_b$ is our main proxy and takes the value of one if bank b is located outside Germany (and zero otherwise). In column 2, the positive and significant coefficient of 0.532 for the interaction term $EU/UN\ sanctions_{c,t} \times Abroad_b$ means that German banks abroad increase their positions in sanctioned countries relative to banks located inside Germany.

Figure 2 illustrates the heterogeneous effects of sanctions on banks located inside and outside Germany. Panel A of Figure 2 shows the cumulative (log) growth of banks' external positions between 12 months before and 18 months after the imposition of financial sanctions. The solid line represents banks that are located in Germany whereas the dashed line represents German bank affiliates abroad. A similar graph is shown in panel B of Figure 2, except that the vertical axis shows (log) assets conditioned on bank $\times$ time and bank $\times$ counterparty country fixed effects. Hence, panel B is equivalent to the regression in column 2 of Table 4.²²

²² To calculate the conditional estimates on the vertical axis, we specify β_S and β_T in Equation (1) as vectors with one element for each of the 12 months before and the 18 months after the imposition of financial sanctions.

Table 4
The effect of financial sanctions on lending across bank locations

	Dependent variable: $\text{Log assets}_{b,c,t}$									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$EU/UN \text{ sanctions}_{c,t}$	-0.275* (0.164)	-0.470*** (0.069)			-0.317* (0.161)			-0.287* (0.161)		
$EU/UN \text{ sanctions}_{c,t} \times \text{Abroad}_b$		0.532*** (0.127)	0.511*** (0.169)	0.761*** (0.239)						
$EU/UN \text{ sanctions}_{c,t} \times \text{Outside FATF}_b$					0.836** (0.372)	0.730** (0.321)	0.340** (0.171)			
$EU/UN \text{ sanctions}_{c,t} \times \text{FATF blacklist}_b$								1.104** (0.461)	1.111** (0.440)	0.708 (0.469)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Counterparty country $\times$ time FE	No	No	Yes	No	No	Yes	No	No	Yes	No
Banking group $\times$ c. country $\times$ time FE	No	No	No	Yes	No	No	Yes	No	No	Yes
Obs.	2,377,900	2,377,900	2,377,900	1,439,900	2,377,900	2,377,900	1,439,900	2,377,900	2,377,900	1,439,900
R^2	.840	.840	.844	.878	.840	.844	.878	.833	.836	.878

This table shows coefficient estimates from panel regressions. The dependent variable $\text{Log assets}_{b,c,t}$ is the natural logarithm of one plus all external assets of bank b with counterparties in country c in month t . The regressor $EU/UN \text{ sanctions}_{c,t}$ is an indicator variable that equals one if Germany is imposing a financial sanction on behalf of the EU or UN on country c in month t (and zero otherwise). The regressors Abroad_b , Outside FATF_b , and FATF Blacklist_b are indicator variables that equal one if bank b is located respectively outside Germany, outside the FATF, or in a country that has been declared noncooperative by the FATF (and zero otherwise). Specifications control for bank $\times$ counterparty country, bank $\times$ time, counterparty country $\times$ time, and banking group $\times$ counterparty country $\times$ time fixed effects as indicated. The sample contains bank-counterparty country-month observations for years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, external position of banks (AUSTA), March 2002 to December 2015, and authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

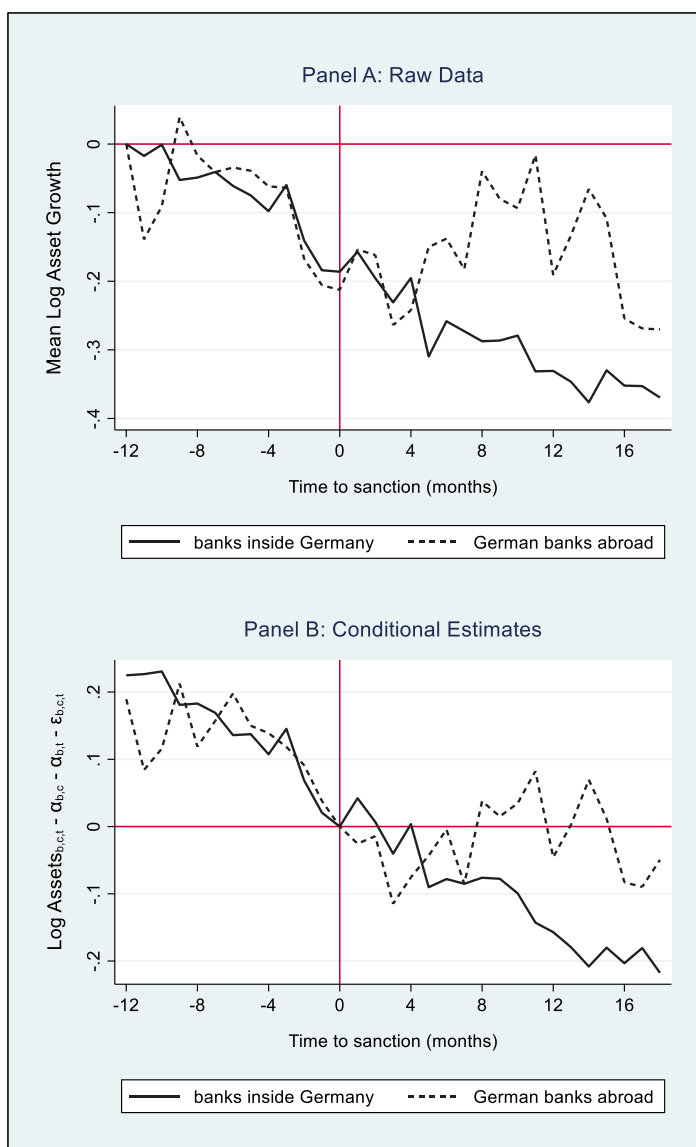


Figure 2

Banks' external positions around the introduction of financial sanctions This figure shows the evolution of banks' external positions around the imposition of financial sanctions. The horizontal axes show event time measured in months. The vertical axis in panel A shows cumulative log asset growth $\text{Log assets}_{b,c}^t - \text{Log assets}_{b,c}^{-12}$ of the average bank since the year before the imposition of financial sanctions. The vertical axis in panel B shows estimated external positions conditionally on bank-time ($\alpha_{b,t}$) and bank-counterparty country fixed effects ($\alpha_{b,c}$). The positions are estimated based on the regression in Equation (1) and calculated relative to the time when financial sanctions are imposed (event time = 0). The solid lines in panels A and B represent banks that are located inside Germany whereas the dashed lines represent German bank affiliates located abroad. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations.

Both panels of Figure 2 show that bank positions decrease already in the year prior to the imposition of financial sanctions. In part, this downward trend could be explained by political developments (human rights violations, war, civic unrest, etc.) that finally lead to the imposition of sanctions and likely depress the demand for credit. Importantly, the general contraction in credit is similar for both groups of banks before the imposition of financial sanctions. As the parallel trends assumption seems to hold, we consider banks in Germany as an appropriate counterfactual for German bank affiliates abroad. Four months after the imposition of sanctions, banks outside Germany increase their positions relative to banks in Germany, which is consistent with the prediction that banks in Germany face a stronger increase in litigation and compliance costs (Hypothesis 1).

To disentangle credit demand from credit supply, we reestimate Equation (1) with counterparty country $\times$ month fixed effects, which absorb all time variation at the country level including, but not limited to, changes in aggregate credit demand (Khwaja and Mian 2008). The results in Table 4, column 3 show a significant effect of 0.511. This estimate is almost identical to the one in column 2, indicating that the relative shift in lending toward banks outside Germany is largely unrelated to factors on the side of borrower countries, but seems to be mostly driven by the banks themselves (or by their institutional environment).

In columns 2 and 3, we do not distinguish between shifts in credit supply that take place across or within German banking groups. In column 4, we therefore include banking group $\times$ counterparty country $\times$ time fixed effects $\alpha_{g,c,t}$, which control for variation across different banking groups, and focus on the group-internal allocation of bank business in sanctioned countries. After dropping singleton observations, we retain 1,439,900 observations. The coefficient for $EU/UN\ sanctions_{c,t} \times Abroad_b$ remains significant and increases in magnitude, suggesting that German bank affiliates abroad increase their positions not only relative to competitors but also relative to other banks that belong to the same group.

Columns 5 to 7 replace $Abroad_b$ by the variable $Outside\ FATF_b$, which is our second proxy for bank locations with weak institutional quality. $Outside\ FATF_b$ equals one if bank b is located in a host country that is not a member state of the FATF (and zero otherwise), thereby classifying countries more cleanly by the strength and commitment of their law enforcement agencies to combat financial crime (see Section 2.1). In column 5, the interaction term $EU/UN\ sanctions_{c,t} \times Outside\ FATF_b$ equals 0.836, suggesting that sanctions shift credit supply toward banks in non-FATF countries. Again, the coefficient changes little when we include counterparty country $\times$ time fixed effects in column 6, and it remains statistically significant when we include banking group $\times$ counterparty country $\times$ time fixed effects in column 7.

A closer comparison of the results reported in columns 2 and 5 illustrates the advantages and disadvantages of using either $Abroad_b$ or $Outside\ FATF_b$.

The coefficient of $EU/UN\ sanctions_{c,t} \times Abroad_b$ is more precisely estimated, with a t -statistic of 4.19, whereas $EU/UN\ sanctions_{c,t} \times Outside\ FATF_b$ has a t -statistic of only 2.25. As shown in Section 2.3, the variable $Abroad_b$ creates less of an imbalance between treatment and control group, accounting for a similar number of observations and similar average position size as domestic banks at home, whereas $Outside\ FATF_b$ identifies a relatively small treatment group (see Table 2).²³ On the other hand, $Outside\ FATF_b$ reduces classification error and makes the treatment group more homogeneous with respect to the quality of institutions and anticrime policies. The differences-in-differences estimate in column 5 is larger than the one in column 2. Calculating the magnitude $exp(\widehat{\beta}_S + \widehat{\beta}_T) - exp(\widehat{\beta}_S)$ of the incremental effect, we find that the positions of banks abroad increase by 43.9% ($= exp(-0.470 + 0.532) - exp(-0.470)$) relative to the positions of domestic banks. By contrast, the positions of banks outside the FATF increase by 95.2% relative to their counterfactual after the imposition of financial sanctions.

These incremental effects are economically significant. According to panel B of Table 2, German banks abroad hold positions in sanctioned countries with an average size of approximately EUR 32m. Relative to this number, an incremental effect of 43.9% implies a sizeable increase by roughly EUR 14m per position. As banks outside the FATF hold much smaller positions in sanctioned countries of, on average, only EUR 4.8m, the large incremental effect of 95.2% translates into an absolute increase by only EUR 4.6m per position.²⁴

Columns 8 to 10 use our last proxy: $FATF\ blacklist_b$. This variable takes the value of one for the nine host countries of German bank affiliates that the FATF has officially declared as noncompliant with respect to its anticrime policies (see Section 2.1). Unsurprisingly, the coefficients of $EU/UN\ sanctions_{c,t} \times FATF\ blacklist_b$ tend to be larger than the interaction terms with either $Abroad_b$ or $Outside\ FATF_b$.

Overall, all three approaches to classifying bank locations provide estimates that are consistent with Hypothesis 1. After Germany imposes sanctions on behalf of the EU or UN, German banks in weak institutional environments increase their positions in targeted countries relative to other German banks, and this shift in the origin of German credit is also taking place within banking groups. Table A.16 in Internet Appendix 4 provides additional evidence that, during the years 2010 to 2015, lending in sanctioned countries by German bank affiliates abroad is positively correlated with intragroup credit that these

²³ See also Figure 1. At the end of our sample period, German banks outside the FATF account for only 2.6% of total German bank assets in sanctioned countries.

²⁴ To provide further context for the incremental effects of 43.9% and 95.2%, we recall that German bank assets in the average sanctioned country grow by 24.3% per month with a large volatility of 183.3% (Table 3, panel C). Values are more moderate for the median sanctioned country, which has an asset growth rate of exactly zero but still a high asset growth volatility of 36.9%.

affiliates receive from their parent banks at home.²⁵ Finally, it is important to recall that, despite the relative increase in lending by some German banks, the average German position (estimated across all bank locations) still decreases by 24% after the imposition of sanctions (see Table 4, column 1).

4.2 Formal enactment versus enforcement of financial sanctions

According to Hypothesis 1, German bank affiliates abroad, and especially outside the FATF, face lower litigation and compliance costs for business in sanctioned countries. Hypothesis 2 suggests two possible sources for this competitive advantage. First, German bank affiliates might be located in host countries that do not enact the sanctions that Germany imposes on behalf of the EU or UN themselves. Hence, these affiliates would not be burdened with lending restrictions and new due diligence requirements in their host countries. Another explanation is that, although bank host countries enact the sanctions *formally*, their relatively low institutional quality inhibits their ability or willingness to enforce them. Economically, lack of formal enactment of sanctions and lack of enforcement both reduce compliance and litigation costs and affect banks' trade-offs in sanctioned countries in the same way. However, from a policy perspective, it is important to understand to what extent formally identical laws and regulations are sufficient to ensure a level playing field across bank locations and to what extent further harmonization of their enforcement is needed.

We begin the analysis by splitting the regression sample into three subsamples:

1. Banks located in Germany
2. Bank affiliates located in foreign host countries that enact EU/UN sanctions themselves
3. Bank affiliates located in foreign host countries that do not enact EU/UN sanctions

The sorting into subsamples takes place at the bank-counterparty country level ($b; c$). For example, bank b 's host country could decide to follow Germany's example and enact UN sanctions against Iran but decide against enacting EU sanctions against Russia.²⁶ In other words, a bank host country may enact some of the sanctions listed in Table 1, but not others. None of the three subsamples includes observations with counterparty countries against which Germany never imposes financial sanctions.

²⁵ We do not observe intragroup credit in years before 2010.

²⁶ Hence, (b ; Iran) would be in sample 2, whereas (b ; Russia) would be in sample 3. In principle, the sorting also would be time-varying because a given bank host country might enact the sanctions against counterparty country c not at the same time as Germany but only with some delay. To ensure that each combination ($b; c$) belongs to only one subsample, we sort all monthly observations ($b; c; t$) into subsample 2 whenever bank b 's host country enacts sanctions against a given country c at some point during our sample period.

Table 5
Formal enactment versus enforcement of financial sanctions: Subsample regressions

Dependent variable: $\text{Log assets}_{b,c,t}$	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Regression sample:</u>							
Bank host country:	Germany	Foreign	Foreign	Inside FATF	Inside FATF	Outside FATF	Outside FATF
Host country enacts EU/UN sanctions:	Yes	Yes	No	Yes	No	Yes	No
$\text{EU/UN sanctions}_{c,t}$	-0.291* (0.144)	-0.032 (0.184)	0.725* (0.339)	-0.236** (0.096)	0.783* (0.435)	0.460** (0.177)	0.426*** (0.099)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	90,759	42,466	25,610	119,011	28,611	5,549	5,663
No. of banking groups	163	39	51	144	126	9	19
No. of individual banks	163	155	243	266	289	31	67
R^2	.85	.75	.83	.82	.85	.79	.75
<u>Coefficient comparison:</u>							
Column 1 vs. column 6	Diff 0.751	SE (0.265)	t -stat 2.84	Confidence interval [0.174; 1.328]			
Column 4 vs. column 6	0.696	(0.303)	2.30	[0.036; 1.357]			

This table shows coefficient estimates from panel regressions with bank $\times$ counterparty country fixed effects. The variables $\text{Log assets}_{b,c,t}$ and $\text{EU/UN sanctions}_{c,t}$ are defined as in Table 4. In columns 1 to 3, we split our data set into three subsamples based on banks' host countries: Germany, foreign bank host countries that enact EU/UN sanctions themselves, and foreign bank host countries that do not enact EU/UN sanctions like Germany. In columns 4 to 7, the data set is split into four subsamples depending on whether bank host countries are inside or outside the FATF and whether they enact or do not enact EU/UN sanctions like Germany. In all columns, the regression samples only include observations with counterparty countries against which Germany imposes EU/UN sanctions at least once between 2002 and 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month in columns 2 to 7 and by counterparty country and month in column 1. To calculate the differences between the coefficients in columns 1 and 4 and column 6 and their statistical significance, we pool the relevant subsamples and estimate regressions with an interaction term between $\text{EU/UN sanctions}_{c,t}$ and a dummy variable for the subsample in column 6. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5 shows separate regressions for each of the three subsamples. For now, we keep these regressions as parsimonious as possible and only control for bank $\times$ counterparty country fixed effects. Column 1 reports the effect of sanctions on banks that are located in Germany. Columns 2 and 3 show the corresponding effects for German bank affiliates that are located in foreign host countries that either enact the EU or UN sanctions themselves (sample 2) or do not (sample 3). The subsample analyses show that banks located in Germany reduce their positions after the imposition of sanctions, whereas German bank affiliates abroad either do not change their positions (column 2) or even increase them if they are located in host countries that do not enact the EU/UN sanctions that Germany is imposing (column 3).

We then extend the subsample analysis by further distinguishing between FATF members and non-FATF members among banks' host countries. Hence, we split the sample into four subsamples depending on whether a host country formally enacts sanctions like Germany and whether it belongs to the coalition of FATF countries with high institutional quality and strong policies against financial crime. According to Table 5, German banks (bank affiliates) only reduce their positions in countries sanctioned by Germany if their host countries formally enact the same sanctions and are also members of the FATF

(column 4). German bank affiliates in all other host countries increase their positions (columns 5 to 7). Importantly, German affiliates increase lending even if their host countries follow the German example and formally enact sanctions themselves but lack the institutions and policies of FATF member states to also enforce them: column 6 shows a large coefficient of 0.460 for banks in sanctioning host countries outside the FATF, which is statistically significantly larger than the coefficients for banks in Germany (column 1) and banks in other in sanctioning host countries inside the FATF (column 4). Hence, Table 5 provides initial evidence that differences in the quality of law enforcement explain why banks in host countries that all enact sanctions at least formally do not adjust their lending in the same way.

We keep the subsample regressions in Table 5 deliberately simple, without complicated interactions, and without bank $\times$ time or counterparty country $\times$ time fixed effects. However, to address the endogeneity concerns discussed in Section 3, we also estimate a more complex difference-in-differences model in the pooled sample:

$$\begin{aligned} \text{Log assets}_{b,c,t} = & \beta_1 \text{EU/UN sanctions}_{c,t} \\ & + \beta_2 \text{EU/UN sanctions}_{c,t} \times \text{Bank host country is foreign} \\ & \text{and enacts EU/UN sanctions}_{b,c,t} \\ & + \beta_3 \text{EU/UN sanctions}_{c,t} \times \text{Bank host country is foreign} \\ & \text{and does not enact EU/UN sanctions}_{b,c,t} \\ & + \alpha_{b,c} + \alpha_{b,t} + \varepsilon_{b,c,t}, \end{aligned} \quad (2)$$

where β_1 measures the base effect of sanctions for banks that are located in Germany. The coefficient β_2 measures the incremental effect for German bank affiliates in foreign host countries that themselves enact the EU/UN sanctions that Germany imposes against counterparty country c in month t . Similarly, β_3 measures the incremental effect for affiliates in host countries that do not enact the EU/UN sanctions imposed by Germany.²⁷ We include bank $\times$ time fixed effects and will later add counterparty country $\times$ time fixed effects, too. The base levels of *Bank host country is foreign and enacts EU/UN sanctions* $_{b,c,t}$ and *Bank host country is foreign and does not enact EU/UN sanctions* $_{b,c,t}$ are collinear with the regressor *EU/UN sanctions* $_{b,c,t}$ and the bank $\times$ counterparty country fixed effects and, hence, omitted.²⁸

²⁷ The distinction between three different bank locations is similar to the one in Table 5, columns 1 to 3. However, in Table 5, a host country that enacts sanctions with some delay relative to Germany is sorted into the category “enacts sanctions” in all months, whereas the model in Equation (2) allows for time-varying categories.

²⁸ We only consider EU and UN financial sanctions, which Germany all imposes as part of its international obligations (see Section 1.1). Hence, if Germany does not enact any EU/UN sanctions against country c (*EU/UN*

Table 6
Formal enactment versus enforcement of financial sanctions: Pooled regressions

Dependent variable: $\text{Log assets}_{b,c,t}$	(1)	(2)	(3)	(4)
$\text{EU/UN sanctions}_{c,t}$	-0.470*** (0.068)		-0.363*** (0.135)	
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is foreign and enacts EU/UN sanctions}_{b,c,t}$	0.444*** (0.123)	0.432*** (0.155)		
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is foreign and does not enact EU/UN sanctions}_{b,c,t}$	0.846*** (0.253)	0.803** (0.306)		
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is inside the FATF and does not enact EU/UN sanctions}_{b,c,t}$			0.664** (0.294)	0.627* (0.336)
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is outside the FATF and enacts EU/UN sanctions}_{b,c,t}$			0.915** (0.375)	0.792** (0.329)
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is outside the FATF and does not enact EU/UN sanctions}_{b,c,t}$			0.801** (0.378)	0.737** (0.330)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes
Counterparty country $\times$ time FE	No	Yes	No	Yes
Obs.	2,377,900	2,377,900	2,377,900	2,377,900
R^2	.840	.844	.840	.844

This table shows coefficient estimates from panel regressions. The variables $\text{Log assets}_{b,c,t}$ and $\text{EU/UN sanctions}_{c,t}$ are defined as in Table 4. In columns 1 and 2, the indicator variable *Bank host country is foreign and enacts EU/UN sanctions* $_{b,c,t}$ (respectively, *Bank host country is foreign and does not enact EU/UN sanctions* $_{b,c,t}$) equals one for bank locations abroad that have enacted (respectively, have not enacted) the EU/UN sanctions against counterparty country c by time t (and zero otherwise). Similarly, in columns 3 and 4, the indicator variables *Bank host country is inside the FATF and does not enact EU/UN sanctions* $_{b,c,t}$, *Bank host country is outside the FATF and enacts EU/UN sanctions* $_{b,c,t}$, and *Bank host country is outside the FATF and does not enact EU/UN sanctions* $_{b,c,t}$ distinguish between bank locations inside and outside the FATF that either have or have not enacted the EU/UN sanctions against counterparty country c by time t (and zero otherwise). Specifications control for bank $\times$ counterparty country, bank $\times$ time, and counterparty country $\times$ time fixed effects as indicated. The sample contains bank-counterparty country-month observations for years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month. Sources: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

Column 1 of Table 6 shows the coefficient estimates for the regression model in Equation (2). Several results are noteworthy. First, the coefficient estimate of -0.470 for $\text{EU/UN sanctions}_{c,t}$ is identical to the one in column 2 of Table 4. Banks in Germany decrease lending after Germany imposes sanctions on behalf of the EU or UN. Second, the incremental effect of 0.846 on lending from affiliates in host countries that do not enact the EU/UN sanctions themselves is statistically significant and corresponds to a large relative increase of 83% ($= \exp(-0.470 + 0.846) - \exp(-0.470)$). Third, banks located in foreign host countries that enact the sanctions like Germany exhibit a roughly 50% lower incremental effect of 0.444 , indicating that local enactment of sanctions indeed helps reduce competitive distortions. Still, the incremental effect corresponds to a sizeable increase of 35% relative to banks in Germany. Fourth, including counterparty country $\times$ time fixed effects does not change the

$\text{Sanctions}_{c,t} = 0$), then there do not exist any EU/UN sanctions against country c , and *Bank host country is foreign and enacts EU/UN sanctions* $_{b,c,t} = 0$ for all $(b; c; t)$. Similarly, $\text{EU/UN Sanctions}_{c,t} = 0$ implies that *Bank host country is foreign and does not enact EU/UN sanctions* $_{b,c,t} = 1$ for all bank affiliates located abroad and 0 for all banks in Germany. Finally, bank $\times$ counterparty country fixed effects absorb whether a bank is in Germany or abroad.

results (column 2). Qualitatively, these conclusions are similar to those drawn from the simple subsample regressions in Table 5, columns 1 to 3. However, the point estimates differ quantitatively (e.g., for banks in Germany). We show in Internet Appendix 3 that the different magnitudes in Tables 5 and 6 are mostly because of the lack of appropriate fixed effects in Table 5.

In column 3 of Table 6, we reestimate the difference-in-differences specification but distinguish between four groups of banks: the first group comprises banks and bank affiliates that are located in Germany or in another FATF member country that enacts sanctions; the second group includes German affiliates in countries that also belong to the FATF but that do not enact sanctions like Germany; and the remaining two groups comprise German bank affiliates located in non-FATF countries that either enact or do not enact the EU/UN sanctions.

We find that German banks in FATF member states that enact sanctions reduce their positions in targeted countries significantly (-0.363). This is not surprising as FATF countries have strict anticrime policies and compliance standards and show a generally high ability and willingness to enforce regulations and policies against financial crime. Similarly, it is intuitive that banks in FATF and non-FATF countries show positive incremental effects of 0.664 and 0.801 if these host countries decide against imposing sanctions themselves. However, it is striking that the incremental effect is also large and significant for bank locations that do enact sanctions formally but that are not members of the FATF (0.915). In fact, the point estimates of 0.915 and 0.801 for sanctioning and non-sanctioning-host bank locations outside the FATF are very similar, which shows that formal enactment of sanctions does not substantially affect bank lending from countries that lack strong institutions and anticrime measures. These results are robust when we control for counterparty country $\times$ time fixed effects in column 4.

4.3 The effect of financial sanctions on bank branches versus subsidiaries

As explained in Sections 1.1 and 1.3, German branches are required to terminate business that is illegal under EU sanctions even if they are located outside the EU. From the perspective of a German parent bank, which is liable for its branches, business in sanctioned countries generates lower litigation risk if conducted by the bank's foreign subsidiaries than by its foreign branches. As sanctions typically ban lending only to some, but not to all, borrowers in targeted countries, this does not imply that branches will terminate all business. However, *ceteris paribus*, the shift in German credit supply toward bank host countries that do not enact or enforce sanctions themselves should be stronger for the foreign subsidiaries of German banks.

In Table 7, we split the sample between foreign subsidiaries in columns 1 and 2 and foreign branches of German banks in columns 3, 4, and 5 and analyze both subsamples separately. We drop all observations that correspond to banks

Table 7
The effect of financial sanctions on bank branches versus subsidiaries

Dependent variable: $\text{Log assets}_{b,c,t}$	(1)	(2)	(3)	(4)	(5)
Regression sample:	German bank subsidiaries abroad		German bank branches abroad		
$EU/UN \text{ sanctions}_{c,t}$	−0.021 (0.015)	−0.141** (0.058)	−0.018 (0.411)	−0.037 (0.438)	−0.037 (0.438)
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country does not enact } EU/UN \text{ sanctions}_{b,c,t}$	0.542*** (0.187)		0.332 (0.364)		
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is inside the FATF and does not enact } EU/UN \text{ sanctions}_{b,c,t}$		0.652*** (0.221)		0.355 (0.391)	0.355 (0.391)
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF and enacts } EU/UN \text{ sanctions}_{b,c,t}$		0.909* (0.503)		confid'l	
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF and does not enact } EU/UN \text{ sanctions}_{b,c,t}$		0.761* (0.438)		confid'l	
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF}_b$					0.225 (0.438)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes	Yes
Obs.	562,864	562,864	597,325	597,325	597,325
R^2	.827	.827	.810	.810	.810

This table shows coefficient estimates from panel regressions. In columns 1 and 2, we restrict the regression sample to German bank subsidiaries located abroad. In columns 3, 4, and 5, the sample comprises only German bank branches abroad. The dependent variable $\text{Log assets}_{b,c,t}$ and the regressors are defined as in Tables 4 and 6. In column 4, data confidentiality requirements prevent us from showing the coefficient estimates of $EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF and enacts } EU/UN \text{ sanctions}_{b,c,t}$ and $EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF and does not enact } EU/UN \text{ sanctions}_{b,c,t}$ because each of them individually is identified by the branches of less than five German banking groups. All specifications control for bank $\times$ counterparty country and bank $\times$ time fixed effects. The sample contains bank-counterparty country-month observations for the years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

domiciled inside Germany because our data do not report the legal status of domestic banks. The two samples of foreign subsidiaries and branches are of similar size and each account for roughly 600,000 bank-counterparty country-month observations.²⁹ Column 1 of Table 6 reports a regression specification similar to column 1 of Table 6. However, the coefficient of $EU/UN \text{ sanctions}_{c,t}$ does not measure the effect of sanctions on banks in Germany (which are missing in Table 7); instead, it measures the effect on German subsidiaries in foreign host countries that are enacting EU/UN sanctions. We find that the estimate of −0.021 for $EU/UN \text{ sanctions}_{c,t}$ is close to zero and insignificant, possibly because it is partly based on host countries that formally enact but do not fully enforce sanctions. The interaction term $EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country does not enact } EU/UN \text{ sanctions}_{b,c,t}$ in column 1 has a significant coefficient of 0.542. Subsidiaries lend relatively more in sanctioned countries if they operate from host countries that do not enact the sanctions that Germany imposes on behalf of the EU or UN.

²⁹ Table 7 focuses on the full regression model in differences with bank $\times$ counterparty country, bank $\times$ time, and counterparty country $\times$ time fixed effects. We also estimate simpler (but less-well-identified) subsample regressions in the spirit of Table 5 for subsidiaries and branches separately (see Internet Appendix 4, Table A.12).

Column 2 of Table 7 distinguishes between the subsidiaries' host countries by FATF membership and whether or not the host countries enact sanctions. The negative coefficient estimate of -0.141 for $EU/UN\ sanctions_{c,t}$ is now statistically significant. German subsidiaries in FATF member states that enact sanctions decrease their positions in the targeted countries. The positive and significant interaction terms in column 2 show that subsidiaries lend more in countries targeted by EU/UN sanctions if they operate from host countries that do not enact the sanctions themselves or if they operate from countries that lack the institutional strength of an FATF member state to enforce the sanctions through effective anticrime policies.

When we consider branches instead of subsidiaries in column 3 of Table 7, the coefficient of the interaction $EU/UN\ sanctions_{c,t} \times Bank\ host\ country\ does\ not\ enact\ EU/UN\ sanctions_{b,c,t}$ decreases relative to column 1 and becomes statistically insignificant. Hence, whether a host country enacts sanctions itself seems to influence the lending decisions of branches less than the lending decisions of subsidiaries, which is consistent with Hypothesis 3. At the same time, the total effect on branches in non-sanctioning-host countries remains large at $0.314 (=0.332 - 0.018)$ and exceeds the negative lending response of -0.470 by banks that are located in Germany (Table 4, column 2).³⁰ In other words, although branches in non-sanctioning-host countries do not increase lending as aggressively as subsidiaries, they do not decrease their positions like the domestic banks at home either.

One possible explanation for why German bank branches abroad do not decrease their positions by as much as their parent banks at home could relate to especially high compliance costs in Germany. As explained in Section 1.2, survey evidence from 898 financial institutions in 16 different countries reveals that Germany, together with the United Kingdom, tops the list of countries with the highest expenses for financial crime compliance worldwide, because of relatively strict documentation requirements, high labor costs, and various other factors like, for example, legal limitations on the use of personal data (LexisNexis 2020).³¹ As a result, some business might generate compliance costs that are too high for banks in Germany, but not for German branches abroad. A second possible explanation could relate to internal governance problems in banking groups. For example, German parent banks might be unable to monitor each individual transaction by their branches abroad.³²

In column 4 of Table 7, we also distinguish between branch host countries by FATF membership and whether or not the host countries enact sanctions. We find that all regressors (and interactions) in column 4 are statistically

³⁰ In unreported regressions, the difference of $0.784 (=0.314 - (-0.470))$ is statistically significant.

³¹ Total annual compliance costs per country were projected to equal US\$47.5bn in Germany but only US\$26.4bn in the United States in 2020 (LexisNexis 2020).

³² For example, the CEO of Danske Bank blamed the money laundering scandal in an Estonian branch on deficiencies in governance and control (Binham and Milne 2018).

insignificant; however, data confidentiality requirements prevent us from reporting the separate point estimates for branches in sanctioning and non-sanctioning-host countries outside the FATF, because each of them individually is identified by the branches of less than five German banking groups. Instead, we can only report one single point estimate for all branches outside the FATF together (without distinguishing further between sanctioning and nonsanctioning-host locations). In column 5, this single interaction term $EU/UN\ sanctions_{c,t} \times Bank\ host\ country\ is\ outside\ the\ FATF_b$ is statistically insignificant, and its coefficient of 0.225 is three to four times smaller than the coefficients of 0.909 and 0.761 in column 2 for subsidiaries outside the FATF. Hence, the quality of institutions and anticrime measures seems to influence branches' lending decisions less than those of subsidiaries.

5. Robustness

In Section 3, we acknowledged the fact that differences in institutional quality across bank locations are not random. Especially non-FATF countries are often different from Germany along many other dimensions besides weaker policies against financial crime. We show that our results are not confounded by cultural ties between sanctioned countries and bank locations outside the FATF (Section 5.1) nor by weaker *financial* regulations (as opposed to anticrime regulations) or by general differences in banks' lending behavior (Section 5.2).

5.1 Cultural, linguistic, and geographic proximity

The international trade literature shows that geographic, linguistic, and cultural proximity can reduce informational asymmetries between countries and facilitate trade (e.g., [Melitz and Toubal 2014](#)). This prediction could be especially relevant in our setting. For example, banks in Germany could find it difficult to ascertain which transactions in (distant) sanctioned countries remain legal and which counterparties were blacklisted. Similarly, banks located very far from the sanctioned countries might find it difficult to judge the economic impact of sanctions on the expected profitability and risk of bank business. Banks in non-FATF countries, on the other hand, could possess local knowledge in sanctioned countries and have “people on the ground” with access to soft information that is unavailable in Germany.

In Table 8, we distinguish between German banks located inside and outside the FATF. Additionally, we include proxies for possible informational asymmetries between bank locations and counterparty countries. In column 1, we interact the dummy $EU/UN\ sanctions_{c,t}$ with the variable *Same Official language* $_{b,c}$, which equals one if counterparty country c and the host country of bank b share the same official language (and zero otherwise). Consistent with the idea that a shared official language facilitates bank business in sanctioned countries, the coefficient of $EU/UN\ sanctions_{c,t} \times Same\ official\ language_{b,c}$ is positive. More importantly, also the interaction term EU/UN

Table 8
Controlling for cultural, linguistic, and geographic proximity

Dependent variable: $\log assets_{b,c,t}$	(1)	(2)	(3)	(4)
$EU/UN\ sanctions_{c,t}$	-0.354** (0.151)	-0.462** (0.187)	-0.373** (0.175)	-0.317* (0.160)
$EU/UN\ sanctions_{c,t} \times Outside\ FATF_b$	0.839** (0.362)	0.799** (0.362)	0.826** (0.368)	0.803** (0.393)
$EU/UN\ sanctions_{c,t} \times Same\ official\ language_{b,c}$	0.677** (0.323)			
$EU/UN\ sanctions_{c,t} \times Spoken\ language\ overlap_{b,c}$		1.389*** (0.407)		
$EU/UN\ sanctions_{c,t} \times Native\ language\ overlap_{b,c}$			4.121*** (1.299)	
$EU/UN\ sanctions_{c,t} \times In\ neighbor\ country_{b,c}$				0.283 (0.497)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes
Obs.	2,312,126	2,262,616	2,262,616	2,377,900
R^2	.840	.835	.835	.840

This table shows coefficient estimates from panel regressions. $\log assets_{b,c,t}$, $EU/UN\ sanctions_{c,t}$, and $Outside\ FATF_b$ are defined as in Table 4. $Same\ official\ language_{b,c}$ is an indicator variable that equals one if bank b is located in a host country that has the same official language as counterparty country c . $Spoken\ language\ overlap_{b,c}$ and $Native\ language\ overlap_{b,c}$ measure the probabilities that two people from bank b 's host country and from counterparty country c speak at least one common language or, respectively, share the same native tongue (Melitz and Toubal 2014). $In\ neighbor\ country_{b,c}$ is an indicator variables that equals one if the host country of bank b and counterparty country c share a border. Specifications control for bank $\times$ counterparty country and bank $\times$ time fixed effects. The sample contains bank-counterparty country-month observations for the years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

$sanctions_{c,t} \times Outside\ FATF_b$ remains significant and its magnitude barely changes compared to the original specification in column 5 of Table 4. In columns 2 and 3, we include interactions with $Spoken\ language\ overlap_{b,c}$ and $Native\ language\ overlap_{b,c}$.³³ These variables measure whether people from two countries can communicate directly without need of translation and proxy for ethnic ties, which can help create trust and facilitate the exchange of soft information (Melitz and Toubal 2014).³⁴ Again, we find that countries' commitment to enforce financial sanctions, as measured by FATF membership, remains important and that the coefficient of $EU/UN\ sanctions_{c,t} \times Outside\ FATF_b$ changes very little.

In column 4, we consider geographic rather than linguistic or ethnic distance, motivated by existing research on relationship lending: "if a lender has to have direct contact with the small business to collect information about it, it has to have a local presence. If, in addition, much of this information is soft and

³³ $Spoken\ language\ overlap_{b,c}$ and $Native\ language\ overlap_{b,c}$ are defined as the probability that two random people from a given country pair speak at least one common language or, respectively, the same native language. We thank Melitz and Toubal (2014) for sharing the data on <http://fariditoubal.com>.

³⁴ $Same\ official\ language_{b,c}$ overstates the probability that people from different countries can communicate with each other. For example, the probability that two random people from South Africa and Liberia understand one another equals only 24% although English is an official language in both countries.

difficult to communicate, the decision to offer credit has to be made very close to where the information is gathered” (Petersen and Rajan 2002, pp. 2533–4).³⁵ In column 4, we interact $EU/UN\ sanctions_{c,t}$ with $In\ neighbor\ country_{b,c}$, which equals one if bank b is located in a country that shares a border with counterparty country c . The insignificant coefficient of this interaction cannot confirm that geographic proximity becomes more important after the imposition of sanctions – unlike the distinction between FATF and non-FATF countries. Clearly, differences in the commitment to enforce financial sanctions across bank locations inside and outside the FATF constitute a distinct channel through which sanctions affect bank lending.

5.2 Financial regulation and supervision

According to Hypothesis 1, weaker anticrime policies and sanctions requirements in non-FATF countries reduce banks’ expected litigation and compliance costs of business in sanctioned countries. However, it is possible that banks in non-FATF countries also benefit from other competitive advantages. For example, bank regulation and supervision could be weaker in general and not just with respect to financial crime.

This raises the question which factor is of greater importance for bank lending in sanctioned countries: weaker overall regulation or a weaker enforcement of anticrime policies and, hence, lower sanction-related costs specifically. In principle, both factors could explain why lending in sanctioned countries shifts to banks outside the FATF. After a negative shock to credit demand, firms should first reduce their borrowing from the most expensive banks. Hence, when sanctions reduce demand, the more heavily regulated and, hence, less competitive banks inside the FATF should lose business first. For this mechanism, it does not matter whether regulatory costs inside the FATF are higher in general or only due to stricter sanctions requirements specifically. The share of total lending coming from the banks inside the FATF would decrease either way.³⁶

We perform two robustness tests to examine this alternative hypothesis. The first one allows for the possibility that banks that face weaker regulations and supervision in general respond differently to sanctions. In column 1 of Table 9, we interact $EU/UN\ sanctions_{c,t}$ with the variable $Official\ supervisory\ power_{b,t} < P25$, which identifies the 25% bank locations with the weakest bank supervisors. $Official\ supervisory\ power_{b,t}$ is a binary variable constructed by Barth, Caprio, and Levine (2013) and not specific to financial crime compliance. For example, it captures the supervisor’s power to order banks to change their loss provisioning scheme, their organizational structure, their management, or their payout policies.

³⁵ See also, for example, Berger et al. (2003), Degryse and Ongena (2005), Buch (2005), and Beck, Ongena, and Sendeniz-Yuncu (2019).

³⁶ We thank an anonymous referee for pointing out this alternative hypothesis.

Table 9
Controlling for the intensity of financial regulation and for banks' ex ante lending behavior

Dependent variable: $\text{Log assets}_{b,c,t}$	(1)	(2)	(3)	(4)	(5)
$\text{EU/UN sanctions}_{c,t}$	-0.320** (0.161)	-0.343** (0.146)	-0.314* (0.162)	-0.163 (0.161)	-0.544** (0.245)
$\text{EU/UN sanctions}_{c,t} \times \text{Outside FATF}_b$	0.840** (0.373)	0.694** (0.331)	0.849** (0.373)	1.040*** (0.338)	1.001** (0.438)
Controlling for the intensity of financial regulation:					
$\text{EU/UN sanctions}_{c,t} \times \text{Official supervisory power}_{b,t} < P25$	0.338* (0.200)				
$\text{EU/UN sanctions}_{c,t} \times \text{Capital regulatory index}_{b,t} < P25$		0.639 (0.396)			
$\text{EU/UN sanctions}_{c,t} \times \text{External governance index}_{b,t} < P25$			-0.137 (0.101)		
Controlling for banks' ex ante lending behavior outside sanction episodes:					
$\text{EU/UN sanctions}_{c,t} \times \text{Lending sensitivity}_b$ (Agg. lending, Agg. borrower-GDP)				0.065 (0.245)	
$\text{EU/UN sanctions}_{c,t} \times \text{Lending sensitivity}_b$ (Agg. lending, Home GDP of b)				-0.578** (0.245)	
$\text{EU/UN sanctions}_{c,t} \times \text{Lending sensitivity}_{b,c}$ (Lending in c, GDP of c)					1.687*** (0.180)
$\text{EU/UN sanctions}_{c,t} \times \text{Lending sensitivity}_{b,c}$ (Lending in c, Home GDP of b)					-0.423 (0.283)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes	Yes
Obs.	2,377,900	2,377,900	2,377,900	2,251,318	1,507,272
R^2	.840	.840	.840	.845	.822

This table shows coefficient estimates from panel regressions. $\text{Log assets}_{b,c,t}$, $\text{EU/UN sanctions}_{c,t}$, and Outside FATF_b are defined as in Table 4. In columns 1 to 3, we interact $\text{EU/UN sanctions}_{c,t}$ with three binary variables that are equal to one for bank locations with weak $\text{Official supervisory power}_{b,t}$, $\text{Capital regulatory index}_{b,t}$, and $\text{External governance index}_{b,t}$ below the 25th percentile (and zero otherwise). The three variables are indexes that Barth, Caprio, and Levine (2013) calculate based on the World Bank surveys on bank regulation. In columns 4 and 5, we interact $\text{EU/UN sanctions}_{c,t}$ with different proxies $\text{Lending sensitivity}$ for the ex ante cyclicity of bank lending outside sanctions episodes. $\text{Lending sensitivity}_b$ (Aggregate lending, Aggregate borrower-GDP) is calculated as the correlation between bank b 's total positions in non-sanctioned-host countries and these countries' aggregate GDP. $\text{Lending sensitivity}_b$ (Aggregate lending, Home GDP of b) is the correlation between bank b 's total positions in non-sanctioned-host countries and the GDP of bank b 's own host country. $\text{Lending sensitivity}_{b,c}$ (Lending in c, GDP of c) is the correlation between bank b 's positions in country c and country c 's GDP. $\text{Lending sensitivity}_{b,c}$ (Lending in c, Home GDP of b) is the correlation between bank b 's positions in country c and the GDP of bank b 's host country. All four variations of $\text{Lending sensitivity}$ are only calculated based on observations from before the imposition of any financial sanctions. Specifications control for bank $\times$ counterparty country, and bank $\times$ time fixed effects. The sample contains bank-counterparty country-month observations for the years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

The positive interaction $\text{EU/UN sanctions}_{c,t} \times \text{Official supervisory power}_{b,t} < P25$ suggests that banks with weak financial supervision lend relatively more after the imposition of sanctions.³⁷ More importantly, the interaction of interest, $\text{EU/UN sanctions}_{c,t} \times \text{Outside FATF}_b$, remains positive

³⁷ The interaction effect $\text{EU/UN sanctions}_{c,t} \times \text{Official supervisory power}_{b,t}$ is significant only when we focus on the bank locations with the weakest $\text{Official supervisory power}_{b,t}$. In unreported regressions, the interaction becomes insignificant when we replace the cutoff at the 25th percentile with, for example, the median.

and significant.³⁸ It seems that the shift in lending toward countries with weaker anticrime policies and weaker law enforcement is not confounded by the intensity of *financial* supervision. This interpretation is also supported by columns 2 and 3, which include interactions with two other variables of Barth, Caprio, and Levine (2013) and consider the stringency of capital requirements ($Capital\ regulatory\ index_{b,t} < P25$) as well as regulatory requirements regarding external audits, bank ratings, and the transparency of financial statements ($External\ governance\ index_{b,t} < P25$), respectively. Again, the interaction term $EU/UN\ sanctions_{c,t} \times Outside\ FATF_b$ remains economically and statistically significant.³⁹

For our second robustness check, we take a closer look at the alternative hypothesis that *general* and not *sanction-specific* differences in competitiveness explain our findings. We note that, if competitive differences are *not* sanction specific, then *any* negative shock to aggregate credit demand should decrease the share of total borrowing that comes from the less competitive and, hence, more expensive banks.⁴⁰ Even outside sanction episodes, the less competitive banks would be the marginal lenders that lose and regain business first when aggregate demand changes. Differences in banks' ex ante lending behavior before the imposition of sanctions should capture any general competitive differences that are unrelated to sanctions. In columns 4 and 5 of Table 9, we control for interactions between the variable $Sanction_{c,t}$ and different measures of banks' ex ante lending behavior outside sanction episodes.

In column 4, the variable $Lending\ sensitivity_b$ (*Aggregated lending, Aggregated borrower GDP*) is computed as the correlation between bank b 's total external positions in non-sanctioned-host countries and these countries' aggregated gross domestic product (GDP). If credit demand is correlated with economic activity, this variable should proxy the demand sensitivity of bank lending.⁴¹ For completeness, we also consider $Lending\ sensitivity_b$ (*Aggregated lending, Home GDP of b*), which is the correlation between bank b 's total external positions in non-sanctioned-host countries and the GDP of bank b 's own host country. We find that the coefficient of $Sanction_{c,t} \times Lending\ sensitivity_b$ (*Agg. lending, Agg. borrower GDP*) is insignificant and does not explain the effect of sanctions on bank lending. The negative and significant coefficient of $Sanction_{c,t} \times Lending\ sensitivity_b$ (*Agg. lending, Home GDP of b*)

³⁸ The correlation between $Outside\ FATF_b$ and $Official\ supervisory\ power_{b,t} < P25$ is only -0.024 . Countries with weak anticrime policies are not necessarily the countries with the weakest financial supervision.

³⁹ Our results are also robust when we interact $EU/UN\ sanctions_{c,t}$ with *Supervisory forbearance discretion*, *Private monitoring index*, or *Overall restrictions on banking activities* (Barth, Caprio, and Levine 2013).

⁴⁰ In Table A.14 in Internet Appendix 4, we find no evidence that this is true. Outside sanctions episodes, changes in a country's total borrowing do not explain changes in the share of credit coming from banks outside the FATF.

⁴¹ The GDP data are reported at quarterly frequency. To calculate the correlation, we only consider bank positions and economic output (GDP) in countries that are never sanctioned or that are not sanctioned yet.

suggests that those banks whose lending is more sensitive to economic output in their host country lend less after sanctions are imposed.

In column 5, we refine the calculation of lending sensitivities and make them counterparty country specific. *Lending Sensitivity_{b,c}* (*Lending in c, GDP of c*) is calculated as the correlation between bank *b*'s positions only in counterparty country *c* and the GDP of country *c*. *Lending Sensitivity_{b,c}* (*Lending in c, Home GDP of b*) is the correlation between bank *b*'s positions in country *c* and the GDP in bank *b*'s own host country. Both variables are calculated based on only observations before any sanctions are imposed on country *c*. We find that the interaction term $EU/UN\ sanctions_{c,t} \times Lending\ sensitivity_{b,c}$ (*Lending in c, GDP of c*) has a positive and significant coefficient whereas $EU/UN\ sanctions_{c,t} \times Lending\ sensitivity_{b,c}$ (*Lending in c, Home GDP of b*) is insignificant now. More importantly, however, the interaction of interest, $EU/UN\ sanctions_{c,t} \times Outside\ FATF_b$, remains statistically significant in column 5. Comparing Tables 4 and 9, we find that the relative increase in lending from banks outside the FATF is quantitatively robust when we control for general differences in regulation (columns 1 to 3) or for ex ante differences in banks' lending behavior (columns 4 and 5).

Internet Appendix 4 reports additional analyses of presanction lending behavior.⁴² In Table A.13, we reestimate Equation (1) in a synthetic sample, which we construct based on the overlap weights method of Li, Morgan, and Zaslavsky (2018) and which balances banks inside and outside the FATF in terms of their lending behavior in non-sanctioned-host countries. In Table A.14, we show that, outside sanction episodes, changes in a country's total borrowing do not affect the fraction of credit that the country borrows from banks outside the FATF. Similarly, idiosyncratic demand shocks, which we calculate at the country level based on Amiti and Weinstein (2018), do not change the fraction of borrowing from banks outside the FATF.

5.3 Other robustness tests

We provide several additional robustness tests in the Internet Appendix. In Table A.2, we exclude the year 2002 from the regression sample because data for 2002 (but not in later years) are left-censored. Table A.4 shows robustness when we use the inverse hyperbolic sine transformation and not the logarithmic transformation of the dependent variable. Tables A.1, A.3, and A.5 show several regressions that use as dependent variables: (log) loans or nonloan assets (A.1), positions with bank or nonbank counterparties (A.3), and only Euro-denominated (nonconverted) positions (A.5). In Table A.6, we show regressions when we exclude the five largest German banking groups from the regression sample. In Table A.7, we use the Worldwide

⁴² We thank an anonymous associate editor for suggesting these tests.

Governance Indicators, which are published by the World Bank Group (Kaufmann, Kraay, and Mastruzzi 2009), and classify bank locations by dimensions like political stability and accountability, governance effectiveness, or corruption. Table A.8 distinguishes between sanctions that are imposed on Russia and Iran and sanctions that are imposed on other countries. In Table A.9, we distinguish between laxer and more intense sanctions as well as between sanctions with a more or less crippling effect on the targeted economy. In Tables A.13 and A.14, we report the additional analyses of banks' presanction lending behavior that were mentioned in Section 5.2. In Table A.15, we show how the standard errors of coefficient estimates change when we cluster them differently.

6. Conclusion

Financial markets are highly integrated, but regulation and supervision are typically local. To curb regulatory distortions to international bank competition, supervisors work together to harmonize rules and financial standards across countries. These efforts include the regulation of financial risks (e.g., Basel I–III regulations), as well as legal risks related to financial crime (FATF standards). Our analysis of financial sanctions and cross-border lending shows that only harmonizing the rules and regulations does not guarantee a level playing field for bank competition. Countries further need to harmonize their ability and willingness to enforce these rules through adequate anticrime measures and strong institutions.

We highlight two questions for future research, which we cannot answer conclusively because of data limitations. First, we are unable to identify who makes the decision in a banking group about whether group-affiliated establishments outside the FATF should lend more in sanctioned countries. For example, it is possible that each bank affiliate decides independently whether or not to do business in sanctioned countries. Headquarters could be simply unaware of such behavior or unable to control their affiliates, for example, because of informational asymmetries. When the CEO of Danske Bank resigned in 2018, he blamed the money laundering scandal in an Estonian branch on “a series of major deficiencies” in governance and control (Binham and Milne 2018). However, there could also exist alternative explanations. Headquarters themselves could be managing and optimizing the allocation of legal risk and compliance costs to their affiliates outside the FATF.

Second, we are unable to distinguish whether banks in weak jurisdictions engage in illegal business with blacklisted counterparties or in legal business with nontargeted entities in sanctioned countries. Illegal business would be more attractive to banks in weak jurisdictions because expected litigation costs should be lower there. However, legal business also would be more attractive to banks in weak jurisdictions because they probably face fewer compliance

requirements with regard to due diligence and recordkeeping. Clearly, this distinction matters from a legal point of view. However, from a business perspective, litigation and compliance costs both have the same effect on a bank's economic trade-off. In each case, banks in weak jurisdictions benefit from a competitive advantage relative to banks in stricter jurisdictions. Cross-country differences in litigation risk as well as in compliance costs both distort bank competition and can trigger evasion behavior.

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